Oncology, DSD, Gender & Ageing

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 24	Basic Principles of Immunology and Immunotherapy in Urologic Oncology
Ch 25	Molecular Genetics and Cancer Biology
Ch 55	Management of Abnormalities of the External Genitalia in Males
Ch 58	Management of Abnormalities of the Genitalia in Girls
Ch 59	Differences of Sexual Development
Ch 60	Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations
Ch 65	Adolescent and Transitional Urology
Ch 120	Aging and Geriatric Urology
Ch 126	Management and Considerations in Transgender and Nonbinary Individuals
Ch 127	Male-to-Female Gender-Affirming Surgery: Vaginoplasty
Ch 128	Special Urologic Considerations in Transmasculine Individuals
★	Putting it together

24 Basic Principles of Immunology and Immunotherapy in Urologic Oncology

1 · Innate and adaptive immunity

- P2 Describe use and principles of immunology and Immunotherapy in Urological Oncology — *innate/adaptive arms, TLR/PAMP, two signals, Th1 vs Th2, checkpoints, adaptive immune resistance, three Es*
- BCG cell-wall peptidoglycans (PAMPs) → **TLR2** on bladder innate cells.
- Urothelial cell = sentinel: type I IFN (↑ class I MHC), CXCL8 (IL-8) = powerful neutrophil attractant.
- **Treg: CD4 subset, marker FoxP3**. Th1 (IL-2, TNF-α, IFN-γ) → **BCG success**; Th2 (IL-4, IL-5, IL-10, IL-13) → **BCG failure**. Raised serum IL-2 after BCG = favourable.

ALGORITHM

Danger signal (TNF-α, TLR ligand) → DC stops antigen uptake → ↑ MHC + B7-1/B7-2 → afferent lymphatics (CCL21 sensed by CCR7) → activates CD4, CD8, B cells in node

	Class I MHC	Class II MHC
Recognised by	CD8	CD4
Distribution	Almost all cells; ↑ by inflammation, IFN	APCs
Peptide	~9 amino acids	13–25 amino acids

2 · Immune editing and escape

- Escape: ↓ tumour antigen, ↓ MHC, ↑ Tregs. Equilibrium may last a lifetime.
- Adaptive immune resistance: CD8 IFN-γ → PD-L1 on tumour/myeloid cells → PD-1 on same CD8 cells → switched off.

KEY POINT

PD-L1 is largely induced by the immune attack itself — a marker of an ongoing but suppressed response, not a fixed tumour trait.

T-cell molecule	Ligand	Effect
CD28	CD80/CD86 (B7)	Activating
CTLA-4	CD80/CD86	Inhibitory; mainly FoxP3+ Tregs
PD-1	PD-L1 (B7-H1), PD-L2	Inhibitory; mainly CD8 TILs
LAG-3	MHC class II	Inhibitory

3 · Inflammation and GU tumours

	Bladder	Kidney	Prostate
Inflammation → cancer	Strong: *S. haematobium* (squamous), chronic UTI, catheters, cyclophosphamide	Weak (smoking, obesity)	Probable: peripheral-zone inflammation; PIA beside HGPIN
CD8 infiltration	Better outcome (MIBC)	**Worse — renal paradox**	May be suppressive

	Bladder	Kidney	Prostate
Dominant restraint	PD-L1, ↑ with grade	PD-1/LAG-3-high T cells, few APCs	Tregs; PD-L1-negative tumour

- Chronic Th2/Treg/M2 inflammation promotes cancer; acute Th1/CD8 eliminates it.

4 · BCG in bladder cancer

ALGORITHM

Instil 1–5 × 10^8 bacilli → adhere via **fibronectin attachment protein** → TLR2/TLR4 on urothelium → IL-8 + TNF-α → neutrophil wave (hours) → DC activation → CD4 Th1 in nodes (weeks) → memory

- Memory implied by ~**50%** of NMIBC in durable remission.
- **Nadofaragene firadenovec**: IFN-α2b protein not durable → non-replicating adenovirus with IFN-α2b gene + **Syn3** (overcomes antiadherence barrier). 3-monthly up to 12 months; CIS CR 53.4%; FDA 2022, **first gene therapy** for BCG-unresponsive NMIBC with CIS.

5 · Cancer vaccines

- P4 Rationale of using cancer vaccines in Urology — *antigen + GM-CSF adjuvant; sipuleucel-T (PAP–GM-CSF, 3 doses 2-weekly, IMPACT); failed IMA901, AGS-003, PROSTVAC*
- Sipuleucel-T: OS **+4.1 months**, **no effect on time to progression**; first FDA-approved solid-tumour vaccine.

6 · Checkpoint blockade

Agent	Target
Nivolumab, pembrolizumab	PD-1
Avelumab, atezolizumab, durvalumab	PD-L1
Ipilimumab	CTLA-4

Bladder

- First-line pembrolizumab alone (KEYNOTE-052): only cisplatin-ineligible with CPS **≥10**, or any-platinum-ineligible; CPS <10 did worse than carboplatin.
- Atezolizumab, durvalumab withdrawn (failed confirmatory trials); adjuvant atezolizumab (IMvigor010) negative.
- **Enfortumab vedotin**: ADC, anti-nectin-4 + monomethyl auristatin E. EV + pembrolizumab ORR 68% cisplatin-ineligible (EV-103). †EV-302: ~doubled PFS and OS vs platinum → first line regardless of cisplatin eligibility.

Kidney

- CTLA-4 on Tregs + PD-1 on CD8 TILs → combined blockade synergistic. VEGF impairs DC maturation and T-cell entry; TKI normalises vessels → rationale for TKI + IO.

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
KEYNOTE-426	Pembrolizumab + axitinib	60% vs 40%	15.4 vs 11.1
JAVELIN Renal 101	Avelumab + axitinib (only anti-PD-L1)	52.5% vs 27.3%	13.3 vs 8.0
CLEAR	Pembrolizumab + lenvatinib (best)	71% vs 36%	23.9 vs 9.2

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
CheckMate 9ER	Nivolumab + cabozantinib	55.7% vs 27.1%	16.6 vs 8.3

Prostate

- Pembrolizumab in mCRPC: PSA response <10% (tumour rarely PD-L1+). Ipilimumab: two phase III, no OS benefit. Atezolizumab + enzalutamide (IMbassador250; enzalutamide ↑ PD-L1): no OS benefit.
- Exception: **MSI-H/dMMR** (1.2–12%) → pembrolizumab second line or beyond; 2017 tumour-agnostic approval — **first based on biomarker, not site**.
- †irAEs (colitis, hepatitis, pneumonitis, endocrinopathies): hold + corticosteroids for grade ≥2; endocrinopathies need lifelong replacement.

> **KEY POINT**
> Every successful GU immunotherapy either supplies a danger signal (BCG, adjuvant, immunogenic cell death, VEGF inhibition) or removes a brake (PD-1, CTLA-4). Failures cluster where the brake is not limiting — PD-L1-negative, Treg-dominated prostate cancer.

7 · Landmark trials

Trial	Design	Result	Take-home
†IMPACT	Sipuleucel-T vs placebo (2:1), mCRPC	OS 25.8 vs 21.7 months	Survival without progression delay
KEYNOTE-057	Pembrolizumab, BCG-unresponsive CIS	CR 41% at 3 months	FDA 2020
KEYNOTE-045	Pembrolizumab vs chemotherapy, post-platinum	OS 10.3 vs 7.4 months	Only anti-PD-1/PD-L1 approved on phase III second-line data
JAVELIN Bladder 100	Avelumab maintenance after platinum	Longer OS	Standard switch maintenance
CheckMate 274	Adjuvant nivolumab vs placebo, high-risk MIBC	DFS 20.8 vs 10.8 months	Adjuvant option
CheckMate 025	Nivolumab vs everolimus, mRCC after TKI	Longer OS, less toxicity	First RCC checkpoint approval
CheckMate 214	Nivolumab + ipilimumab vs sunitinib, first line	OS HR 0.63	First line intermediate/poor risk
KEYNOTE-564	Adjuvant pembrolizumab 1 year vs placebo, high-risk clear-cell RCC	Recurrence/death ↓32%	Adjuvant standard

25 Molecular Genetics and Cancer Biology

1 · Hallmarks and basic molecular genetics

- **Local invasion** is the one hallmark that uniquely separates malignant from benign.
- Pre-mRNA: 5′ capping, 3′ polyadenylation, splicing. **Alternative splicing**: AR splice variants lacking ligand-binding domain (AR-V7) active without ligand → castration resistance.
- **Somatic** testing (tumour tissue): targetable mutations, resistance, heterogeneity. **Germline** (blood/saliva): hereditary syndromes, surveillance, preventive surgery, family cascade.

2 · Gene alterations: tumour suppressors and oncogenes

- P1 Oncogenes — *gain of function/one allele vs suppressor loss/both alleles; mutation, amplification, fusion (TMPRSS2–ERG ~half, MYC, MET); Knudson two-hit*
- Suppressor inactivation: homozygous deletion; loss + mutation; two mutations; loss + promoter methylation.
- Knudson: familial retinoblastoma often **bilateral**. Exceptions: **dominant negative** (TP53 mutant disables wild-type in multimers); **haploinsufficiency** (p27/CDKN1B).

3 · Cell cycle

- G0 → G1 → S → G2 → M; ~24 h, unidirectional. CDKs active only with phase-specific **cyclin** (proteasome-degraded after each phase); also regulated by CDK phosphorylation and CDKIs.

Phase	Complex	Function
G1	Cyclin D–CDK4/6, cyclin E–CDK2	Phosphorylate RB1; cyclin D needs sustained mitogen (Ras/Raf/MEK)
S	Cyclin A–CDK2, then cyclin A–CDK1	S-phase genes
G2/M	Cyclin B–CDK1	Phosphorylates lamins → nuclear envelope breakdown

Checkpoint	Trigger	Mechanism
G1/S	Mitogens, nutrients, O_2, DNA damage	p21 (CDKN1A), 14-3-3σ inactivate G1 CDKs; TP53 → apoptosis if unrepaired
G2/M	DNA damage (radiation)	ATM → CHEK1/CHEK2 → Cdc25 sequestered; TP53 induces p21, GADD45
Spindle assembly	Unattached chromosome	Blocks anaphase-promoting complex; failure → non-disjunction → **aneuploidy** (near-universal in solid tumours)

- CDKIs: **Cip/Kip** (p21, p27, p57) broad; **INK4** (p15, p16, p18, p19) CDK4/6 only → block RB1 phosphorylation.

RB1 — guardian of G1/S

> **ALGORITHM**
> Hypophosphorylated RB1 binds E2F + HDACs → S-phase genes silent → cyclin D–CDK4/6, cyclin E–CDK2 hyperphosphorylate RB1 → E2F released → S-phase genes transcribed → restriction point passed (irreversible)

- Disabled by RB1 loss, cyclin D amplification or CDKI loss — altered in almost all cancers.

- **TP53** (guardian of the genome): DNA damage (ATM/ATR → CHK1/CHK2) or oncogene stress (ARF blocks MDM2) → arrest, repair, senescence or apoptosis. **MDM2** = negative regulator. Most mutated gene in cancer (>half); mediates G1/S and G2/M arrest.

> **KEY POINT**
> Cancer must break both RB1 (entry) and TP53 (integrity) — so both pathways are lost in almost every tumour.

4 · DNA damage and repair

- **Loeb mutator hypothesis**: early loss of replication fidelity → hypermutable tumour selecting resistant clones.

Pathway	Lesion	Key proteins	Cancer link
NER	Bulky adducts, cross-links (UV, chemicals)	XPC, XPD, ERCC1/XPF	Xeroderma pigmentosum; **ERCC2 in bladder cancer**
BER	Single-base damage, single-strand breaks	MUTYH, OGG1, PARP1/2	OGG1 variants → RCC
MMR	Mispairs, indel loops at repeats	MSH2/MSH6, MLH1/PMS2	~100-fold ↑ mutation rate; Lynch
HR	Double-strand break (DSB)	ATM, RAD51, BRCA1, BRCA2	Breast/ovarian, prostate
NHEJ	DSB	Ku70/Ku80, DNA-PKcs, ligase IV	Translocations

- DSB: ATM phosphorylates TP53, CHEK1/2, H2AX → **γH2AX** foci. Unrepaired DSB → apoptosis (basis of radiotherapy, many cytotoxics).
- **HR** uses sister chromatid (only S/G2) → precise. **NHEJ**: no template, any phase → error-prone, translocations.

> **KEY POINT**
> Repair defects are targets: HR loss → PARP inhibitor (synthetic lethality); MMR loss → immunotherapy; ERCC2 loss → cisplatin sensitivity.

5 · Apoptosis

- Apoptosis: programmed, energy-requiring, little inflammation. Caspases: initiators 8, 9, 10 → executioners 3, 6, 7.

	Intrinsic (mitochondrial)	Extrinsic (death receptor)
Trigger	Unrepaired DNA damage, lost survival signals	Fas ligand/TNF on Fas (CD95)/TNFR1
Sequence	Bax/Bak → cytochrome c → Apaf-1 apoptosome → caspase-9	FADD → procaspase-8 → caspase-8
Brakes	Bcl-2, Bcl-XL, Mcl-1; IAPs	FLIP
TP53	Dependent (transactivates Bax)	**Independent** → target when TP53 mutated

- Bid (cleaved by caspase-8) links the two. TP53 loss → chemo/radiotherapy resistance.

6 · Senescence, telomeres, autophagy

- P3 Telomerase — *~100 bp lost/division, two barriers → telomerase immortalisation; normal tissue expression, ALT (ATRX/DAXX); short telomeres in premalignant lesions, predict prostate recurrence*
- Telomerase high in **85–90%** of tumours.
- **Senescence**: viable, permanently arrested (p16, p21, p27); marker SA-β-galactosidase; SASP immunostimulatory but potentially pro-tumour. ADT induces senescence in some prostate cancers; SASP may explain ADT-associated inflammation.
- **Autophagy**: suppresses early tumours (BECN1 loss in prostate cancer), sustains established ones; chloroquine inhibits.

7 · Epigenetics

- **Reversible** change without sequence change → biomarker and drug target.
- **DNA methylation**: 5-methylcytosine at CpG islands (upstream of ~40% of promoters). DNMT1 maintains; DNMT3A/3B de novo. 5-mC deaminates → C→T transitions.
- Promoter hypermethylation silences suppressors ("second hit"): **GSTP1 in >90% of prostate cancer** (commonest somatic change); RASSF1A (up to 97% bladder); p16. Global hypomethylation → instability, oncogene/imprinted re-expression: LINE-1 (bladder), IGF2 loss of imprinting (prostate).
- **Histones**: activating H3K4me3, acetylation; repressive H3K9me3, **H3K27me3 (written by EZH2)**. EZH2 ↑ in mCRPC, predicts PSA recurrence.
- **ncRNA**: long ncRNA >200 nt (HOTAIR → castration resistance; MALAT1). miRNA 18–25 nt silences **many** mRNAs (**miR-371a-3p** testis); siRNA silences **one**.

8 · Hypoxia, VHL and angiogenesis

ALGORITHM

Normoxia: HIF-α prolyl-hydroxylated → bound by VHL (E3 ubiquitin ligase) → proteasomal degradation. Hypoxia or VHL loss: HIF-α accumulates → dimerises with HIF-1β → VEGF, angiopoietin, glycolytic genes

- **Warburg effect**: aerobic glycolysis, best seen in clear cell RCC (constitutive HIF-1α/2α). Hypoxia also drives invasion and immunosuppression (MDSCs, PD-L1, blocked DC maturation).

KEY POINT

VHL loss = permanent pseudohypoxia; angiogenesis, Warburg metabolism, invasion and immunosuppression follow — why clear cell RCC responds to VEGF, HIF-2α and immune blockade.

- **Lineage plasticity**: trans-differentiation under driver targeting (neuroendocrine prostate cancer).

9 · GU tumour genomics

Prostate

Gene	Significance
TMPRSS2–ERG	Commonest fusion (~half)

Gene	Significance
SPOP	Commonest mutated suppressor in primary
TP53	Commonest alteration overall (more in advanced)
RB1, PTEN, TP53	Loss of ≥2 = aggressive variant, overlaps neuroendocrine
BRCA1/2	Germline + somatic hit → **PARP inhibitor**
ATM	Little PARP benefit
CDK12	Tandem duplications; poor response to hormones, chemo, PARP
MMR deficiency	MSI → immunotherapy
AR amplification	Late, under AR-targeted pressure → ADT resistance

Kidney

Subtype	Genomic hallmark	Therapy link
Clear cell (75–80%)	**3p loss** + VHL, PBRM1, BAP1, SETD2	VEGF TKIs; belzutifan (HIF-2α)
Papillary (15–20%)	MET (7q); FH loss in subset	MET-active TKIs
Chromophobe	Mitochondrial DNA, mTOR pathway	mTOR-directed rational
Translocation	MiT/TFE fusions; young women	No targeted therapy
Sarcomatoid (any)	BAP1, CDKN2A loss	PD-L1+ → immunotherapy-responsive

- Clear cell: VHL mutation 60–70% + methylation 5–10%; **PBRM1** (~40%) better; **BAP1**, **SETD2** (~10% each) worse; PBRM1 and BAP1 mutually exclusive.

Bladder (UC)

- High mutation load (APOBEC); TP53 most frequent; KDM6A, ARID1A, FGFR3, PIK3CA, RB1.
- Six consensus subtypes: **basal/squamous largest (35%)**, luminal papillary 24%, neuroendocrine-like smallest 3%.
- **FGFR3** (15–20% advanced; up to 37% upper tract): luminal papillary, T-cell-poor, poor IO response → **erdafitinib**; test all advanced UC. HER2 9–12% of MIBC. ERCC2 → cisplatin sensitivity, bladder-sparing trials.

Testis

- Low mutation rate; **12p gain** (†isochromosome 12p); KIT/KRAS (seminoma); TP53 mutation rare. miR-371a-3p >90% sensitivity and specificity.

Hereditary syndromes

- P2 P1 Hereditary urological malignancies / Familial malignancies in urology — *VHL, HPRC (MET), HLRCC (FH), BHD (FLCN), BRCA2 > BRCA1, Lynch, HOXB13*
- VHL: clear cell RCC >70% lifetime; belzutifan. HLRCC: papillary type 2, collecting duct. Lynch: UC + prostate. HOXB13 → earlier PSA screening.

> **KEY POINT**
> The germline syndrome names the somatic driver (VHL → clear cell, MET → papillary, FH → type 2, BRCA → prostate) and the driver names the drug (HIF-2α, MET TKI, PARP, FGFR inhibitor, immunotherapy for MMR loss).

10 · Clinical NGS and biomarkers

- **Tumour-only** panels cannot separate somatic from germline.

- **VAF** = fraction of reads with variant; depends on zygosity, copy number, LOH, purity (heterozygous × 60% purity → 30%). High = clonal/early; low = subclonal/late; ~50%/~100% → suspect heterozygous/homozygous **germline**; single-digit % in cfDNA → consider **clonal haematopoiesis**.
- **Liquid biopsy**: ctDNA within cfDNA → deep sequencing; post-treatment ctDNA = minimal residual disease.
- Prediction proven only by a **biomarker–treatment interaction test** in an RCT: parallel lines = prognostic, crossing = predictive.
- Absolute benefit scales with baseline risk: 50% relative reduction → 15% absolute at 30% risk, 1.5% at 3%.
- **HSD3B1 (1245A>C)** ↑ DHT synthesis → ADT resistance, greater abiraterone sensitivity.

11 · Landmark trials

Trial	Design	Result	Take-home
†PROfound	Olaparib vs enzalutamide/abiraterone, HR-altered mCRPC	Better rPFS and OS, driven by BRCA1/2	Germline + somatic HR testing in metastatic prostate cancer
†THOR	Erdafitinib vs chemotherapy, FGFR-altered metastatic UC after IO	Longer OS	FGFR testing of all advanced UC

55 Management of Abnormalities of the External Genitalia in Males

1 · Normal development and penile size

- Male differentiation (weeks 9–13): testosterone (fetal Leydig) → DHT (5α-reductase) → functional AR. Any break → undervirilisation; female is the passive default.
- Genital tubercle → glans; urethral folds → shaft; labioscrotal swellings → scrotum (inferomedial migration + fusion = raphe). Glanular urethra probably from fusion of urethral plate, not ectodermal ingrowth.
- **Stretched penile length (SPL)**: pubis (fat pad compressed) to glans tip; correlates with erect length. Term **3.5 ± 0.7 cm** (diameter 1.1 cm); adult 13.3 ± 1.6 cm.
- Micropenis cut-off (−2.5 SD): **1.9 cm** term–5 months; 1.5 cm at 30 weeks, 2.0 cm at 34 weeks; ≈ 3.7–3.9 cm at 5–11 years; adult 9.3 cm. Compare with cut-off, not mean.
- Minipuberty (first 3 months): gonadotrophin surge → second phase of penile growth.
- Tanner: scrotal/testicular change first (stage 2), penile length 3, girth/glans 4. Varicocele peaks Tanner 3; semen analysis at Tanner 5.

2 · Prepuce: phimosis and paraphimosis

- **Physiological phimosis** at birth; smegma and erections separate prepuce: ≈ **90% retractable by 3 years**, < 1% phimotic at 17 years. Forceful retraction → cicatricial (secondary) phimosis.
- Treat for: persistent phimosis, balanitis, posthitis, BXO, UTI.
- **Topical steroid**: betamethasone 0.05% twice daily × 30 days (or thrice daily × 21 days); response within 2 weeks; ≈ 1 in 6 fails (more collagen, less elastic fibre).
- Retractile prepuce halves recurrent febrile UTI risk; steroid for phimosis after UTI (normal US) reduces recurrence.
- Commonest acquired "penile lump": yellowish **entrapped smegma**.
- **Paraphimosis**: oedema within hours → may progress to gangrene. Manual glans compression + distal traction (succeeds in most) → oedema reduction (iced glove 5 min, granulated sugar 1–2 h, punctures) → dorsal slit → circumcision.

3 · Circumcision

- AAP 2012: benefits insufficient to recommend routine neonatal circumcision but justify access and payment.
- Devices (Gomco, Mogen, Plastibell): separate prepuce and inspect meatus first (exclude hypospadias, megameatus). Older children: freehand under GA.
- Analgesia mandatory: **dorsal penile nerve block superior to EMLA** (prilocaine → methaemoglobinaemia); ring block alternative.

Defer / avoid neonatal circumcision	Reason
Hypospadias, curvature, torsion, dorsal hood, buried or webbed penis	Prepuce needed for reconstruction
Large hydrocele / inguinal hernia	Skin pushed forward → secondary phimosis, trapped penis

Defer / avoid neonatal circumcision	Reason
Coagulopathy (contraindication); hyperbilirubinaemia (some)	Haemorrhage
NICU setting	More complications

- **Benefits**: penile cancer almost only in men not circumcised at birth; lower febrile infant UTI (to ≥ 5 years), but **111** circumcisions prevent one UTI; less epididymitis, balanitis; ≈ **60%** protection against HIV acquisition in men (no reduction in transmission to partners); lower HSV-2, high-risk HPV; no proven sexual dysfunction. Neonatal more cost-effective.

Complications

- P1 Circumcision undone — *0.2–5% (10–20× after infancy); bleeding commonest; adhesions/skin bridges (steroids don't help); trapped penis; revision under GA*
- Degloving: re-epithelialisation with ointment + warm baths; **no immediate suturing or graft**.
- Adhesions separate spontaneously (rare after 9 y); office lysis with EMLA. Skin bridge (can cause chordee/torsion) → excise.
- Excess excision → chordee, torsion: flaps, Z-plasty, split-thickness graft. Wound infection rare (Fournier reported).
- Amputated glans: re-attach within **8 h**. Penile necrosis (cautery on clamp): phallic reconstruction vs female reassignment — latter problematic (androgen imprinting).

> **KEY POINT**
> Late sequelae are problems of skin quantity (too much → trapped penis, chordee, torsion; too little → redundancy, adhesions) or of the newly exposed meatus (stenosis).

Meatal stenosis and BXO

- P2 P1 Management of Urethral meatal stenosis / Lichen sclerosus (BXO) — *normal calibre by age, post-circumcision meatitis, upward stream, ventral meatotomy; BXO features, steroid, circumcision, buccal mucosa*
- Meatal stenosis almost exclusively **circumcised** boys (also adhesion lysis, frenular devascularisation, BXO). Urethrography only if UTI, incontinence or proximal anomaly suspected. Suturing mucosa to glans reduces recurrence (sutureless clamp equal).
- BXO in boys: rare < 5 years; steroids only for mild preputial disease; **circumcision preferred** ± meatoplasty; ≈ 1/5 with meatal disease need further meatal surgery. Prepuce-sparing: triradiate preputioplasty + intralesional triamcinolone.

4 · Inconspicuous penis versus micropenis

> **KEY POINT**
> **Stretched length is the single discriminator.** Inconspicuous penis (buried, webbed, trapped) = normal SPL and diameter hidden by skin, fat or scar → anatomical treatment. Micropenis = small but normally formed → endocrine work-up.

Buried penis type	Mechanism	Treatment
Congenital	Inelastic dartos; skin not anchored to deep fascia	Surgery
Obesity	Suprapubic fat hides shaft	Weight loss; liposuction only in adolescents

Buried penis type	Mechanism	Treatment
Trapped (post-circumcision)	Cicatrix embeds penis in fat	Dilation or betamethasone + retraction → repair under GA at ≥ 6 months

- Exam: retract skin lateral to shaft — glans exposes easily = normal shaft. Infant fat pad resolves with walking.
- Trapped penis risks: voiding difficulty, pooling → balanitis, UTI; psychosocial.
- **Repair**: excise dysgenic dartos bands (proximal dorsal) → unfurl prepuce for ventral cover → fix dorsal subcutaneous tissue to pubic fascia and scrotal tissue to ventral base. Skin: Z-plasty, preputial island flap, graft.
- **Webbed penis** (penoscrotal fusion): congenital or after excess ventral skin removal; transverse incision + vertical closure, or Byars' flap. Avoid neonatal circumcision.

Micropenis

- Normally formed, ≥ 2.5 SD below mean (< **1.9 cm** at term); normal length:girth ratio. Testes often small/cryptorchid.
- Differentiation complete by week 12 under placental hCG (defect → small penis *with hypospadias*); later growth needs fetal LH → gonadotrophin deficiency after 14 weeks → normally formed micropenis.
- Chromosomal: Klinefelter, X polysomies, trisomies 8, 13, 18.

Cause	Examples	Hormones
Hypogonadotrophic hypogonadism (commonest)	Kallmann, Prader–Willi, Laurence–Moon–Biedl, CHARGE, hypopituitarism (hypoglycaemia, GH deficiency), midline brain defects	Low LH/FSH
Hypergonadotrophic	Rudimentary testes, gonadal dysgenesis, Robinow	High LH/FSH; no testosterone rise after hCG
Idiopathic	Delayed gonadotrophin timing	Normal axis
Partial androgen insensitivity (rare)	—	Ambiguity commoner

ALGORITHM

History, exam, **karyotype**, SPL → paediatric endocrinologist → testosterone before/after hCG (± GnRH test) → anterior pituitary screen (glucose, Na/K, cortisol, thyroid) → MRI brain → **end-organ response: IM testosterone enanthate × 3 months** (short courses do not affect height; prolonged advance bone age); repeat at puberty.

- Outcome: male identity, standing voiding, satisfactory sexual function in most.

5 · Curvature (chordee without hypospadias) and torsion

- P1 Congenital Penile curvature — *Devine–Horton types, artificial erection, plication/septotomy/graft*
- Repair if sexual dysfunction expected, with concurrent penile surgery, or likely to worsen with growth.

Type	Associations	Repair (stepwise)
Ventral (commonest)	Hypospadias; dorsal hood; prominent frenulum → distal chordee	Deglove → excise fibrous tissue superficial to Buck's → Byars' flaps → dorsal plication/Nesbit/corporal rotation → short urethra: reconstruction; frenular: frenulotomy

Type	Associations	Repair (stepwise)
Dorsal	Isolated, epispadias	Ventral plication; protect dorsal neurovascular bundles
Lateral	Corporal asymmetry; straight when flaccid, noticed on erection	Plicate corpus **opposite** maximal curvature

- Superficial → deep; plicate convex side only if curvature persists. **Artificial erection** locates apex and confirms correction. → *Paediatric Urology book, Ch 56* for hypospadias.
- **Penile torsion**: usually **counter-clockwise (to the left)**, < 60°; skin-level deformity, corpora at base normal. < 90°: deglove, rotate glans opposite, suture to skin. ≥ 90°: mobilise base, divide bands, non-absorbable anchoring sutures at corporal base ± dorsal dartos flap.

6 · Abnormal number, masses and lymphoedema

Feature	Aphallia	Diphallia
Incidence / defect	1 in 10–30 million; genital tubercle fails; 46,XY, normal scrotum + testes	1 in 5 million; accessory to complete duplication
Associations	Cryptorchidism, VUR, renal agenesis, horseshoe, imperforate anus, cardiopulmonary; ≈ 1/3 stillbirth/neonatal death	Hypospadias, bifid scrotum, bladder duplication, renal, pubic diastasis, anal, cardiac
Work-up / decision	Karyotype, anomaly screen, MRI → neophallus vs reassignment (reassigned may develop male identity)	Renal US + VCUG + penile US/MRI → individualised

- Aphallia urethral opening: **postsphincteric** (anal verge) commonest, best survival; presphincteric (prostatorectal fistula) intermediate; urethral atresia + vesicorectal fistula rarest, fatal.
- Penile cysts commonest (parameatal, median raphe, inclusion) → excise; **juvenile xanthogranuloma — observe 1 year**.
- Lymphoedema: Milroy (AD, at birth), Meige (1st–2nd decade), praecox (puberty, ≈ 80%); acquired: anogenital Crohn (anti-TNF). Turner, Noonan, Klinefelter. Surgery excisional (tissue between Buck's and skin), not drainage.
- Haemangiomas involute (propranolol if ulcerated); malformations persist ("bag of worms") → imaging → resection ± embolisation. Klippel–Trénaunay–Weber: genital lesions bleed.

7 · Urethral anomalies

- P2 Megameatus intact prepuce / Anterior urethral valve — *MIP: wide meatus under complete prepuce, found at circumcision, pyramid repair; AUV: bulbar, diverticulum lip, VCUG, endoscopic ablation*
- **Congenital urethral fistula** (subcoronal, normal meatus): multilayer closure, or Thiersch–Duplay if glans bridge thin.
- **Urethral duplication**: sagittal (≈ 94%); **ventral channel functional**. **Effmann**: I incomplete (IA distal, IB proximal blind); II complete (IIA two meatuses — IIA2 "Y-type" perineal; IIB one meatus); III with bladder duplication. Double stream, incontinence (more proximal → more), UTI; VUR commonest association. VCUG, RUG, cystoscopy. **Never use the hypoplastic dorsal channel as main urethra**.
- **Cowper syringocele** (Maizels): simple, perforated (diverticulum-like), imperforate (obstructive cyst), ruptured. Symptomatic → endoscopic unroofing.

- **Megalourethra**: **scaphoid** (commoner; spongiosum deficient, ventral ballooning) vs **fusiform** (rarer; spongiosum + cavernosa deficient, flaccid, often lethal anomalies). Prune-belly, upper-tract anomalies; reduction urethroplasty.

8 · Priapism

- P3 Priapism: Diagnosis and Management — *ischaemic vs non-ischaemic vs stuttering, blood gas, aspiration/phenylephrine, distal → proximal shunts, embolisation*

Type (children)	Cause	Treatment
Ischaemic	**Sickle cell** (commonest), leukaemia	Hydration, O_2, alkalinisation, analgesia ± transfusion → aspiration ± phenylephrine → shunt
Non-ischaemic	Straddle injury → arteriocavernous fistula	Observe → superselective embolisation
Stuttering	Sickle cell	Bedtime pseudoephedrine; oral β-agonist; long-term PDE5 inhibitor
Neonatal	Idiopathic, birth trauma, polycythaemia	None — resolves in 2–6 days

- Corporal blood gas decides urgency: ischaemic = dark, hypoxic, acidotic, rigid, painful (compartment emergency); arterial = bright red, not rigid/painful.
- Sickle: sleep hypoventilation → ↓ corporal pH/O_2 → sickling → stasis → vicious cycle. Counsel about fibrosis and ED.

9 · Scrotal anomalies

- **Penoscrotal transposition**: failed inferomedial migration; partial (shawl, doughnut, prepenile). Associations: perineal/scrotal hypospadias, caudal regression, sex chromosome anomalies, Aarskog–Scott; complete form: **up to 75%** upper-tract anomalies → renal US + VCUG. Repair (with hypospadias): **rotation-flap scrotoplasty**.

Anomaly	Feature	Association	Management
Bifid scrotum	Unfused folds	Proximal hypospadias	Scrotoplasty with hypospadias repair
Ectopic scrotum	Suprainguinal (commonest), infrainguinal, perineal	Ipsilateral renal agenesis/dysplasia, ectopic ureter; cryptorchidism; perineal lipoma usually isolated	Upper-tract US; scrotoplasty + orchidopexy 6–12 months
Hypoplasia / agenesis	Under-developed / absent	UDT; agenesis — all XY, bilateral UDT	—
Scrotoschisis	Rupture, extruded testis	Meconium peritonitis	Explore, orchidopexy, close

10 · Inguinal hernia and hydrocele

All reflect where/how completely the **processus vaginalis** failed to close (≈ 18% closed at term).

Entity	Processus vaginalis	Plan
Indirect hernia	Patent; contents enter	Repair soon after diagnosis

Entity	Processus vaginalis	Plan
Communicating hydrocele	Patent, fluid only	Repair if > 1–2 y, enlarging or fluctuating
Hydrocele of cord	Isolated patent segment	Rarely resolves; repair after 1 year
Scrotal (non-communicating)	Closed	Observe — most resolve by 1 y
Abdominoscrotal	Closed; through ring (≈ 1%; bilateral 30%)	Excise/drain; ring tightening unnecessary

- Hernia 1–5% of children; prematurity (≈ 13% < 32 weeks, ≈ 30% if < 1 kg); right 3:1. Bilateral risk: female, premature, < 1 year, cryptorchidism; ascites/VP shunt.
- **Intermittent bulge** = patent processus → surgery; fixed → observe. Parent photographs reliable; silk-stocking sign. Neonatal bowel transilluminates. US only if obese, incarcerated, doubtful.
- Incarceration → immediate exploration. Reducible → elective within weeks. Preterm: repair before discharge, or delay if very sick/< 1500 g.

> **ALGORITHM**
> Skin-crease incision → open external oblique, spare ilioinguinal nerve (young infants: rings coincide — Mitchell-Banks, not opened) → split cremaster → sac anteromedial → separate from vas/vessels → **high ligation** at internal ring → tighten wide ring → **confirm testis scrotal**.

- Peripubertal non-communicating hydrocele (> 10–12 y) → scrotal approach.
- Recurrence 0.5–1% (2% preterm, 3–6% after incarceration), mostly < 2 years: missed/torn sac, damaged floor, missed direct hernia. Others: iatrogenic cryptorchidism, atrophy (after incarceration), vas injury. Persistent hydrocele after repair → observe 1 year → scrotal repair.
- Laparoscopic repair now matches open; sees and closes contralateral ring.
- **Abdominoscrotal hydrocele**: may cause hydroureteronephrosis, limb oedema; inguinal excision ± orchidopexy.
- Contralateral patency ≈ 40% infants, but metachronous hernia only ≈ **7%** → against routine contralateral exploration; laparoscopy via sac > 99% accurate.

11 · Acute scrotum

Only cord torsion is time-critical. Differential: trauma, hernia, orchitis (mumps, HSP), tumour, **referred pain** (ureteric calculus, appendicitis).

Feature	Cord torsion	Appendage torsion	Epididymitis
Frequency	20–30%	**40–60% (commonest)**	5–15%
Peak age	Perinatal; 12–16 y	7–12 y	Infancy, puberty
Onset / vomiting	Acute severe; vomiting common	Sudden or insidious; uncommon	Insidious; rare
Cremasteric reflex	Usually absent	Present	Present
Signs	High-riding, horizontal lie	**Blue dot**, upper-pole nodule	Tender epididymis; pyuria 20–40%

Feature	Cord torsion	Appendage torsion	Epididymitis
CDUS	Absent/reduced flow; **whirlpool/snail sign**	Avascular upper-pole lesion, hyperaemia	↑ epididymal flow
Management	**Emergency exploration**	Ice, NSAIDs, rest	Supportive; antibiotics only if UTI

Intravaginal torsion

- **Bell-clapper** (≈ 12% at autopsy); familial; cryptorchid at higher risk; left-sided; ≈ 1 in 4000. Delay to presentation is main cause of loss.
- Cremasteric reflex present in up to 10% of torsions — does not exclude.
- **TWIST score**: †swelling 2, hard testis 2, absent reflex 1, nausea/vomiting 1, high-riding 1; ≥ 5 high risk → theatre without imaging; ≤ 2 low risk.
- CDUS: absent flow specific, sensitivity only 63–90%. High-resolution cord US: twist seen in almost all. **Do not delay surgery for imaging when suspicion high.**

Hours from onset	0–6	7–12	13–18	19–24	> 24	> 48
Orchidectomy risk	≈ 5%	20%	40%	60%	80%	90%

- Degree of twist matters (lost ≈ 540° vs 360°); transfer delay lowers salvage.

> **ALGORITHM**
> Suspected torsion → scrotal exploration → detorse, warm packs → **meanwhile fix contralateral testis with non-absorbable suture** → reassess: viable → dartos pouch/fixation (non-absorbable); non-viable → orchidectomy; prosthesis later.

- Manual detorsion fails in ≈ 1/3 (twist opposite to assumed); still needs orchidopexy. Tunica albuginea incision ± tunica vaginalis flap for compartment syndrome — unproven.
- Late: atrophy despite salvage (> 1 day pain, heterogeneity); semen defects, low inhibin B (ischaemia–reperfusion, not antisperm antibodies).
- **Intermittent torsion**: precedes acute torsion in 30–50% → **elective bilateral orchidopexy**.

Perinatal (extravaginal) torsion

- Whole cord + tunica twist before scrotal fixation; hard painless testis, discoloured skin, or vanished nubbin. Salvage rare; bilateral 5–22%. Urgency debated (urgent if after a documented normal postnatal exam); inguinal approach.

Appendage torsion, epididymitis, others

- Appendix testis (Müllerian, ≈ 80% of testes), appendix epididymis (Wolffian). Blue dot may be falsely positive in cord torsion. Explore only if cord torsion not excluded or pain prolonged.
- Epididymitis: ask about UTI, sexual activity, catheterisation, dysfunctional voiding. Ice, NSAIDs, elevation; Gram-negative antibiotics if pyuria; consider STI treatment in adolescents. Prepubertal positive culture → renal US + VCUG (ectopic ureter, urethral/ejaculatory duct anomaly); recurrent → endoscopy.
- Idiopathic scrotal oedema (eosinophilia, **fountain sign**; antihistamine); HSP (may cause torsion); cystic dysplasia of rete testis (ipsilateral renal agenesis).

12 · Adolescent varicocele

- P1 P3 Varicocele – Pathophysiology / Varicocele: controversies — *left-sided anatomy, nutcracker, grading, right/non-decompressing → image for mass; adolescent: observe, repair for asymmetry*

- Prevalence ≈ **15%**; after 10 years, peaks Tanner 3; tall thin boys; familial.
- Testicular volume by **US (L × W × H × 0.71)**; volume doubles Tanner 2→3 → serial measures. Peak retrograde flow > 38 cm/s correlates with asymmetry.
- Surrogates do not reliably predict adult semen; catch-up growth may be spontaneous → **observation default**. Intratesticular varicocele (veins > 2 mm) with asymmetry → repair.

> **KEY POINT**
> **Indications for repair**: left testis > **20%** smaller (or bilateral hypotrophy), pain, abnormal semen (Tanner 5 / ≥ 18 y), intratesticular varicocele with asymmetry. Total volume < 30 mL and low total motile count predict poor semen.

Technique	Recurrence	Hydrocele	Note
Open retroperitoneal (Palomo, mass ligation)	2–4%	Up to 30%	Lymphatics ligated
Laparoscopic, non-sparing	0–9%	11–32%	—
Laparoscopic artery/lymphatic-sparing	1–7%	0–4%	Lymphatic sparing cuts hydrocele
Microscopic subinguinal	0–10%	0–6%	Best balance; rare atrophy
Non-microscopic inguinal	7–33%	8–14%	—
Sclerotherapy (retrograde/antegrade)	6–35% / 4–12%	Occasional	Mainly for recurrence

- Laparoscopic: clip veins accompanying the artery (prevents recurrence). Atrophy risk: artery ligation then later surgery on vasal vessels.
- Hydroceles may appear ≥ 2 years later; up to half need intervention. Genitofemoral nerve injury → thigh paraesthesia.
- Sclerotherapy: retrograde transfemoral (technical failure 5–22%; radiation) or antegrade scrotal; sodium tetradecyl sulfate, polidocanol, coils.

13 · Epididymal and vasal anomalies

- Epididymal cysts: common; spermatoceles postpubertal, contain sperm; mostly observe.
- Absent vas: bilateral → **CFTR**; unilateral → **renal anomalies** (Wolffian).
- Persistent mesonephric duct: vasoureteral fusion + ipsilateral renal and seminal vesicle agenesis → UTI, epididymo-orchitis.
- Microlithiasis, undescended testis → *Paediatric Urology book, Ch 57*.

58 Management of Abnormalities of the Genitalia in Girls

1 · Embryology that explains the anomalies

- **Urorectal septum** divides cloaca into urogenital sinus (ventral) and anorectal canal; node = **perineal body**, splitting cloacal sphincter into urogenital diaphragm and external anal sphincter → **both pudendal-innervated**.
- Müllerian ducts arise lateral to wolffian ducts → **müllerian anomaly + ipsilateral renal anomaly**. Fused ducts → uterus, cervix, upper two-thirds of vagina; **sinovaginal bulbs** → distal vagina. Remnants: Skene (≈ prostate), Bartholin (≈ bulbourethral), Gartner duct (wolffian).
- **Caudal regression**: weeks 4–5 insult; maternal diabetes; earlier insult → higher anorectal malformation. Cranial extension → VATER, MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia).

2 · Evaluation and classification

- Frog-leg; **pull-down manoeuvre** (labia majora drawn caudally/laterally) shows distal vagina. Discharge/bleeding → vaginoscopy under anaesthesia.
- US for anatomy and upper tracts; **MRI = gold standard** for müllerian anatomy. ASRM classification broadest.

Category	Mechanism	Examples
Hypoplasia/agenesis	Fails to form	MRKH
Vertical fusion	Faulty müllerian–sinus canalisation	Imperforate hymen, transverse septum, distal atresia
Lateral fusion	Failed midline fusion/resorption	Didelphys, HWWS

> **KEY POINT**
> Decision hinge: **is it obstructive?** Obstruction needs prompt drainage; non-obstructive anomalies wait until intercourse or fertility demands surgery.

3 · Clitoris

- **Clitoromegaly**: commonest **CAH** → electrolytes, 17-OHP, karyotype → *Ch 59*. Others: maternal androgen tumour, Turner mosaic DSD, plexiform neurofibroma. Normal corpora + large hood = **pseudo-enlargement**.
- **Diminutive clitoris** → CAIS.

4 · Vestibule: the interlabial mass

- **Urethral prolapse**: prepubertal African-ancestry girls, postmenopausal women; **doughnut with central meatus** — no imaging. Sitz baths, topical steroid/oestrogen (recurrence high) → **circumferential excision**; avoid ligation over catheter, cryosurgery.
- **Urethral polyp**: benign; excise. **Skene cyst**: maternal oestrogen; **displaced meatus, eccentric stream**; usually ruptures.

- **Gartner cyst / covered ectopic ureter**: anteromedial vaginal wall; incise; upper-pole heminephrectomy if wetting/infection persists.
- **Prolapsed ureterocele**: infant, retention, pink → dusky; duplex upper pole. Decompress → reduce, usually under GA.
- **Ectopic ureter** below sphincter → **daytime wetting despite normal voiding** (†continuous dribbling); single-system may mimic absent kidney.

> **KEY POINT**
> Read every mass against the meatus: central (prolapse), displaced (Skene), separate/vaginal (Gartner, ectopic ureter, hymen, tumour). Urethral- or vaginal-origin cyst → **always renal US**.

- **Labial swelling**: commonest inguinal hernia. **CALME** (asymmetric labium majus enlargement): physiological → observe; growth → exclude neurofibroma, sarcoma, vascular malformation, Crohn.

5 · Vagina: obstruction and absence

Entity	Level / uterus	Clue	Treatment
Imperforate hymen	Hymen; normal	Whitish bulge (neonate); bluish + amenorrhoea (adolescent)	**Transverse** incision; not aspiration
Transverse septum	Upper/middle third; normal	Normal hymen, blind vagina	Excision, Z-plasty, balloon; hollow mould
Distal atresia	Sinus; normal	Shallow dimple; distended vagina per rectum	Pull-through ± flaps
MRKH	Bulbs/plate; absent	Hymenal fringe + pouch; 46,XX	Dilation first
Cervical agenesis	Cervix; uterus present	Obstruction; MRI	Often hysterectomy + neovagina
HWWS	Lateral fusion; didelphys	Menstruates; lateral bluish bulge	Wide septal incision

- **Imperforate hymen**: probably commonest obstructive anomaly. Transverse incision avoids urethra/rectum; aspiration → ascending infection. Hymenal injury suggests abuse.
- **Transverse septum**: ~1 in 70,000; delay → retrograde menstruation → endometriosis. **MRI distinguishes high septum from cervical agenesis**. GnRH agonist/continuous pill lets lower segment dilate. **Z-plasty** converts transverse scar to longitudinal. Main complication **stenosis**. **Hollow mould** when uterus present.
- **Distal atresia**: haematocolpos acts as tissue expander → drain → pull-through.

MRKH (müllerian agenesis)

- ~1 in 5000; 46,XX, **functioning ovaries**; second commonest primary amenorrhoea (after gonadal dysgenesis). Some have functioning rudimentary uterus (cyclic pain).
- Renal anomaly ≈ **one-third**; skeletal 10–20%; no lumbosacral dysraphism. **Type A**: symmetrical buds, normal tubes, no extra-müllerian anomaly. **Type B**: asymmetrical, renal/Klippel–Feil/MURCS → full renal and skeletal work-up.

CAIS → *Ch 59*; urogenital sinus and cloaca → *Ch 60*.

6 · Vaginal replacement

- **Dilation first-line** (Frank; Ingram bicycle seat): needs dimple and compliance; surgery not superior. **Vecchietti**: olive + laparoscopic traction.
- **McIndoe**: split-thickness graft over stent in urethrorectal space; stenosis → dilation mandatory. Full-thickness graft/Williams labial pouch contract less. **Buccal mucosa**; **robotic peritoneal vaginoplasty** (no bowel, no hair).
- **Sigmoid** (default bowel): excise ~3 cm intervening bowel (no overlapping suture lines → fewer fistulae); fix to posterior peritoneum (prevents prolapse). **Ileum** stenoses more; only for irradiated pelvis/absent colon.
- Bowel: lubricates, less contracture; discharge, **adenocarcinoma**. Skin: stenosis, **squamous carcinoma**, condylomata. All neovaginas → **annual examination**.

> **KEY POINT**
> **Cervical atresia**: perineum-to-uterus anastomosis loses the endocervical barrier → life-threatening ascending infection. Hysterectomy + neovagina often best; otherwise endometrial suppression and IVF with transmyometrial transfer.

7 · Lateral fusion anomalies

- **Didelphys**: septate vagina up to 75%; excise septum only for dyspareunia, delivery obstruction or two tampons. **Septate uterus**: incise only for recurrent pregnancy loss.
- **HWWS**: didelphys + unilateral imperforate hemivagina → pain **without amenorrhoea**; lateral bluish bulge; **ipsilateral renal agenesis**. Wide septal incision. Zhu class 1 complete (earlier), class 2 perforated (commoner, later).

8 · Acquired disorders

- **Vaginal rhabdomyosarcoma**: grapelike bleeding mass, <2 years; best prognosis of genital primaries (embryonal). Biopsy → CT + marrow → chemotherapy ± resection. IRS trials cut surgery from **100% to 13%**.
- **Labial adhesions**: commonest interlabial abnormality; peak 13–23 months; minora fuse (majora fusion → DSD). **80% resolve in 1 year** → treat only symptoms: topical oestrogen (breast budding, pigmentation) or betamethasone 0.05% + pull-down → office separation (EMLA) → surgery. Recurrence depends on **family compliance**.
- P1 Female circumcision — *FGM ~200 million, WHO types I–IV (III = infibulation), complications, midline deinfibulation*
- **Condylomata** (HPV): suspect abuse, but perinatal transmission occurs; >50% resolve; imiquimod, CO_2 laser; vaccination prevents.
- **Inguinal hernia** may present CAIS (~1–2.5%) → uterus on US, cervix, or karyotype; **round ligament absent in genetic males**. If CAIS, preserve testes (aromatised oestrogen).

59 Differences of Sexual Development

1 · Definition and nomenclature

- **DSD**: chromosomal, gonadal or anatomical sex atypical. **2006 Chicago consensus** (Lee) replaced "intersex", "hermaphrodite", "sex reversal" with karyotype-based terms; "atypical" preferred to "ambiguous".

Old term	2006 term
Male pseudohermaphrodite	46,XY DSD
Female pseudohermaphrodite	46,XX DSD
True hermaphrodite	Ovotesticular DSD
XX male / XX sex reversal	46,XX testicular DSD
XY sex reversal	46,XY complete gonadal dysgenesis (not AIS) MCQ

- Offer endocrine/genetic testing for **proximal hypospadias even with descended testes**.

2 · Normal sex development

- Chromosomal sex → gonadal sex (bipotential until ≈ 6 weeks) → phenotypic sex (bipotential until ≈ 8 weeks).
- **SRY** → upregulates **SOX9** → Sertoli. Pleiotropy explains syndromic DSD: **WT1** (Wilms, nephropathy), **SF1/DAX1** (adrenal failure), SOX9 (campomelic dysplasia). DAX1 duplication (Xp21) → 46,XY dysgenesis. Ovary: RSPO1/WNT4/β-catenin, FOXL2.
- **Sertoli → AMH** at 7–8 weeks → Müllerian regression by ≈ 10 weeks. **Leydig → testosterone** from ≈ 9 weeks (autonomous → placental hCG → fetal LH) → Wolffian ducts.
- **5α-reductase 2 → DHT** (higher receptor affinity): active androgen for prostate, urogenital sinus, external genitalia; virilisation complete by 12–13 weeks; third-trimester penile growth and descent (androgen, INSL3).

> **KEY POINT**
> The testis uses two separate hormones: AMH removes Müllerian ducts, testosterone/DHT builds Wolffian and external structures. A uterus in 46,XY means Sertoli (AMH) failure — dysgenesis, not a synthesis or receptor defect.

3 · Classification (2006 / Chicago)

Category	Conditions
Sex chromosome DSD	47,XXY Klinefelter; 45,X Turner; 45,X/46,XY MGD; 46,XX/46,XY chimerism
46,XX DSD — gonadal	Gonadal dysgenesis; ovotesticular DSD; testicular DSD (SRY+, SOX9 duplication)
46,XX — androgen excess	Fetal: CAH (21-OH, 11β-OH, 3β-HSD, P450 oxidoreductase); fetoplacental: aromatase deficiency; maternal: luteoma, drugs
46,XX — other	Cloacal anomalies; MRKH; vaginal atresia
46,XY DSD — gonadal	Complete/partial gonadal dysgenesis (SRY, SOX9, SF1, WT1, DHH, MAP3K1); ovotesticular; testis regression

Category	Conditions
46,XY — androgen synthesis	LH receptor; Smith–Lemli–Opitz; StAR; 3β-HSD; 17α-hydroxylase; 17β-HSD3; 5α-reductase 2
46,XY — androgen action	CAIS, PAIS
46,XY — other	PMDS; vanishing testis; isolated hypospadias; hypogonadotropic hypogonadism; cryptorchidism

- 3β-HSD and P450 oxidoreductase virilise 46,XX yet undervirilise 46,XY. 21-OHD: no genital difference in 46,XY.

4 · 46,XX DSD

CAH — 21-hydroxylase deficiency

- Commonest 46,XX DSD and **commonest cause of neonatal atypical genitalia**. 21-OHD = **95%** of CAH; AR, ≈ 1 in 15,000. Cortisol ↓ → ACTH ↑ → androgen excess; aldosterone ↓ → **salt wasting**.
- 46,XX: normal ovaries and Müllerian structures; often **common urogenital sinus**; may look like a boy with **bilateral impalpable gonads**. **Prader I–V** (†V = male-appearing phallus, meatus at tip, fused empty scrotum).
- **↑17-OHP** after **36 h**; also ↑androstenedione, testosterone, renin; ↓cortisol; cosyntropin test.
- **Salt-wasting crisis** (↓Na, ↑K, acidosis, shock) usually not before **day 5–7**. Glucocorticoid + mineralocorticoid + salt (†hydrocortisone ≈ 10–15 mg/m²/day, fludrocortisone, NaCl in infancy; double-to-triple in illness/surgery).
- 46,XY boys have normal genitalia → **newborn screening** matters most for them.
- Long-term: balance androgen suppression vs steroid excess; "backdoor" androgens make control difficult; fertility reduced. 46,XY: **testicular adrenal rest tumours** — benign, regress with more glucocorticoid, obstruct tubules → bank sperm; excision rarely. Poor suppression → hypogonadotropic hypogonadism.
- Prenatal dexamethasone (start by 6–7 weeks; ≥ 4 fetuses treated per girl helped) — experimental only.
- Forms: classic salt-wasting (crisis in first weeks); simple virilising (no shock); nonclassic (20–50% enzyme activity; premature adrenarche, hirsutism, oligomenorrhoea; may need no treatment).

Enzyme	Share	BP / K^+	46,XY	Signature
21-hydroxylase	95%	Salt wasting; ↓BP, ↑K^+	Normal	↑17-OHP
11β-hydroxylase	≈ 5%	Hypertension, ↓K^+	Normal	—
3β-HSD	<1%	Salt wasting	Undervirilised	↑DHEA
17α-hydroxylase	Rare	Hypertension, ↓K^+	Female/atypical; delayed puberty	↓androgens

Other 46,XX DSD

- P1 P2 46-XX Male — *SRY translocated to X; testes, no Müllerian structures; azoospermia (no AZF)*
- Aromatase (CYP19) deficiency virilises mother and infant; recurs in later pregnancies. Maternal androgen/aromatase exposure ends at birth → no postnatal therapy.

5 · 46,XY DSD

Condition	Key features
Complete gonadal dysgenesis (Swyer)	Streaks, female genitalia, uterus; absent puberty; tumour 30–45% → prophylactic gonadectomy; oestrogen; donor-ovum pregnancy
Partial gonadal dysgenesis	Dysgenetic testes, atypical genitalia, variable Müllerian; WT1 → Denys–Drash (congenital nephrotic, Wilms 90%) / Frasier (late FSGS, gonadoblastoma 60%, no Wilms)
CAIS	X-linked; female genitalia, short blind vagina, no uterus; breasts by aromatisation; ≈ 50% inguinal hernia; 1–2% of girls with hernia are CAIS
PAIS	Atypical genitalia; identity mostly male; pubertal breast growth
5α-reductase 2	AR; atypical/female at birth, blind pouch; **virilises at puberty**, fertility reported; T:DHT **>20:1**; "guevedoces" (Imperato-McGinley 1979)
17β-HSD3	↑androstenedione:testosterone; virilises at puberty
LH receptor defect	Leydig hypoplasia; no hCG response
PMDS	AMH/AMH-receptor mutation (AR); male genitalia + uterus/tubes; found at hernia/orchidopexy; 30–50% of transverse testicular ectopia
Vanishing testes	Loss after first trimester → male genitalia, impalpable testes, no Müllerian; undetectable testosterone/AMH, ↑gonadotropins at mini-puberty; testosterone from adolescence

- Synthesis/action defects: AMH normal → no uterus; adrenal failure only if enzyme shared with cortisol pathway.
- PAIS raised male: high-dose testosterone may help; female: GnRH agonist or gonadectomy then oestrogen. CAIS: dilation usually suffices. Girl at herniorrhaphy: vaginoscopy for cervix or look for testis in sac.
- **PMDS (Clarnette)**: 1 bilateral intra-abdominal (60–70%); 2 testis in sac/scrotum + contralateral hernia (20–30%, "hernia uteri inguinale"); 3 transverse ectopia (10%). Orchidopexy; vasa hug uterus → remnants traditionally kept; Müllerian malignancy 3–8% → removal (laparoscopic, spare vasa) vs imaging surveillance.

6 · Sex chromosome DSD and ovotesticular DSD

- P4 P3 Klinefelter's syndrome / Mixed gonadal dysgenesis — *47,XXY features, hormones, micro-TESE; MGD 45,X/46,XY asymmetry, second commonest atypical genitalia, tumour 15–35%, remove streaks, Turner screening*
- Klinefelter: 1 in 600 males; check gonadotropins/testosterone at mini-puberty; no testosterone in infancy.
- **Turner** (25–50 per 100,000): short, absent puberty, webbed neck, coarctation, bicuspid valve, horseshoe kidney (renal ⅓–½); ovarian insufficiency ≈ 95%. Cardiac/renal screening lifelong; GH; oestrogen + progesterone; oocyte donation or cryopreservation. **Turner + Y**: 12–20% (gonadoblastoma → dysgerminoma); 2017 guideline gonadectomy at diagnosis (now challenged).
- MGD: ≈ 85% of 45,X/46,XY have male-typical genitalia; gender dysphoria in a substantial minority raised female.
- **Ovotesticular DSD**: ovarian + testicular tissue; mostly 46,XX; sinus and uterus usual. **Vas always accompanies a testis**, tube an ovary. Ovary usually normal, testis dysgenetic;

tumour ≈ 3%; polar biopsy for bipolar ovotestis. Defer irreversible steps; puberty usually oestrogenic.

7 · Evaluation of the newborn with atypical genitalia

- P3 DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia — *history, exam, gonads as key sign, karyotype/FISH, 17-OHP, 1-week hormone table, karyotype-based algorithm, sex of rearing, surgery timing, gonadal risk*

> **KEY POINT**
> **Emergency**: atypical genitalia with bilateral impalpable gonads = salt-wasting CAH until proven otherwise. Check electrolytes and 17-OHP; watch for crisis from the end of week 1. Designation and surgery are not emergencies.

- MDT review within **24–48 h**; "your baby", never "it". Urologist without a team → transfer early.
- Gonads: 0 palpable → CAH; 1 → MGD, OVT, PAIS, partial dysgenesis; 2 → 46,XY DSD.
- Hypospadias + UDT (Kaefer 1999): DSD ≈ 30%; 50% if testis impalpable; 65% if proximal meatus.

> **ALGORITHM**
> Karyotype (≥45 cells; FISH X/Y in hours) + electrolytes + 17-OHP at ≈ 36 h → pelvic US (uterus, gonads) → at ≈ 1 week: LH, FSH, testosterone, oestrogen, androstenedione, DHT, AMH ± hCG stimulation → genetic panel/microarray/exome → if unclear: MRI, genitography, endoscopy, laparoscopy ± biopsy (**no gonadectomy during diagnosis**)

- AMH normal → functioning Sertoli (or granulosa) cells; ↑LH/FSH → gonadal failure; DHEA-S ↑ in 21-OHD, 3β-HSD.
- Precise diagnosis nearly always in 46,XX DSD but **<50%** of 46,XY. Late presenters: CAIS, Swyer, MRKH, Turner + Y, Klinefelter, 5α-reductase. After puberty: ↑gonadotropins + ↓sex steroid → gonadal failure; ↑LH + ↑testosterone with female genitalia → CAIS. Examine adolescents with few clinicians, only when management may change.

8 · Gender assignment and MDT

- Team incl. psychologist, ethicist, peer support; **same-day joint clinic**; lifelong care with transition.
- **Money (1975)** model (early designation, early surgery, secrecy) failed → transparency, shared decisions.
- Sex of rearing predicts satisfaction better than androgen exposure; no test predicts identity. Female trend: 46,XX CAH, CAIS, Swyer, LH receptor defect. Male: PAIS, 5α-reductase, 17β-HSD3, partial dysgenesis.
- Dysphoria highest in 5α-reductase deficiency raised female (≈ half); lowest in 46,XX CAH raised female.
- Micropenis (<2.5 SD; <2 cm at 0–5 months) → endocrine work-up; female reassignment no longer recommended.

9 · Surgery

- **Feminising** (clitoroplasty, vaginoplasty/sinus mobilisation, labioplasty → *Ch 60*): clitoral surgery irreversible; infant benefit unproven; only urgent indication urinary

retention/obstruction.

- **Masculinising**: proximal hypospadias ± penoscrotal transposition, orchidopexy; high complications but revisable → infant repair often chosen.
- Ideally defer irreversible surgery until the individual can participate; CAH debated separately.
- MRKH (renal anomaly ≈ ⅓ → renal US): dilation first; uterine remnants removed/suppressed — **never joined to a neovagina** (no cervix); uterus transplantation investigational.

10 · Gonadal tumour risk

Risk ↑ with Y material (gonadoblastoma locus, TSPY), dysgenesis, intra-abdominal position; ↓ with scrotal gonad. Shift to individualised risk, function and autonomy.

Risk	Condition	Germ cell tumour	Management
High	Dysgenesis + Y, intra-abdominal (Swyer, MGD)	15–35% (Swyer 30–45%)	Gonadectomy at diagnosis
	Frasier	60%	Gonadectomy
	Denys–Drash	40%	Gonadectomy
	PAIS, non-scrotal	50%	Gonadectomy
Intermediate	Turner + Y	12%	Gonadectomy (debated)
	17β-HSD3	28%	Monitor or gonadectomy
	PAIS / dysgenesis, scrotal	Unknown	Biopsy at puberty (≥30 tubules, OCT3/4)
Low	CAIS	2%	Keep until after breast development
	Ovotesticular DSD	3%	Remove discordant testicular tissue
?None	5α-reductase; Leydig hypoplasia	Unknown	Unresolved

11 · Ethics

- Bodily autonomy, transparency, no irreversible procedures before consent unless necessary; parental consent covers limits of predicting identity and fertility. Higher psychosocial morbidity → integral psychology; respect preferred terms.

60 Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

1 · Principles

- P3 DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia — *classification, work-up, CAH care, team, sex of rearing, gonadal risk*
- Controversy is **timing and who decides**, not technique. Present pros and cons of surgery *and* no surgery; never persuade; support whichever path; offer all support groups. Diagnosis and gender assignment → *Ch 59*.

2 · Urogenital sinus and cloaca — classification

Urogenital sinus: vagina persistently communicates with the urinary tract (usually mid–distal urethra); both exit as one **common channel**.

Entity	Distinguishing feature
Genital atypia states	Commonest; mostly CAH
Pure urogenital sinus	Normal external genitalia; behaves like a cloaca
Persistent cloaca	Rectum also joins: **one perineal opening**
Female exstrophy	Probably a form of urogenital sinus

- CAH: half of all newborn atypical genitalia, **90%** of 46,XX; salt wasting **67–75%**; karyotype within **48 h**.
- Cloaca: ≈ 1 in 20,000–50,000; 5–13.6% of anorectal malformations.

> KEY POINT
> **Governing principle**: **true urethral length** (bladder neck → confluence) decides the operation far more than common-channel length, in sinus and cloaca. Confluence level is a continuum, not just "high" vs "low".

- **Ganesan concept** MCQ: in CAH the proximal urethra is normal (bladder neck to confluence **1.5–2 cm** in infants); virilisation lengthens **vagina-to-meatus** distance, so "high" CAH confluence is an illusion.
- **Rink classification**: common-channel length, bladder-neck-to-vagina distance, confluence level, clitoral size, appearance.

3 · Clinical evaluation

- **First threat is metabolic**: salt wasting → adrenal crisis; monitor electrolytes before reconstruction. **Hypertension** → 11β-hydroxylase deficiency.
- **Suprapubic mass**: bladder and/or **hydrometrocolpos** — often the only early sign of pure sinus.
- **Huffman index** (glans width × length): normal **<3.5 mm²**, concern **>10 mm²**. Isolated clitoral hypertrophy in a preterm is not a DSD.
- **Both gonads descended → 46,XX extremely unlikely**. Labioscrotal hyperpigmentation in CAH.

- **Posterior cloaca**: sinus opens into the anus (type A) or just anterior (type B).

Hydrocolpos in cloaca: urine refluxes into poorly draining vagina(s) + maternal-oestrogen mucus. Commoner with common channel **>3 cm**, duplication, posterior cloaca. Severe → respiratory distress, cyanotic legs, acidosis.

Associated (cloaca)	Findings	Frequency
Renal	Solitary kidney, dysplasia, PUJ obstruction, ectopia	Majority
Uterovaginal	Duplication/septation	**53–67%**
Spinal	Sacral anomalies; tethered cord ≈ one-third (mostly no skin stigmata)	Cord anomaly on MRI ≤ 43%
VACTERL; GI	TOF; malrotation, duodenal atresia	Correct malrotation at colostomy

KEY POINT

Two early dangers: **adrenal crisis** (CAH) and **obstructive hydrocolpos** (pure sinus, cloaca). Number of perineal orifices tells you which world you are in.

4 · Imaging and endoscopy

Modality	CAH sinus	Pure sinus / cloaca
US	Mandatory; **cerebriform adrenals**	Mandatory for kidneys; poor for Müllerian anatomy
Genitography	Least useful; often inaccurate	**Mandatory**, as CT/MRI cloacogram
MRI	Limited	Pelvis, spine, sphincter muscles
Endoscopy	At reconstruction	Early, separate: decompress and define

- **Genitogram**: **cervical impression** at vaginal dome = female internal organs.
- **Gonadal biopsy** only if it would change gender of rearing; **deep longitudinal incision** (ovarian tissue may surround testicular part of ovotestis).
- **Prenatal cloaca** (from 19 weeks): fetal ascites + bilobed pelvic cysts + hydronephrosis + reduced liquor. **Echocardiography always.**
- **Cloacogram**: catheters in bladder, vagina(s), rectum → CT/MRI 3-D reconstruction: confluences, urethral length, vaginas, true rectum (may lie well above fistula).
- Duplicated vaginas (≈ half) lie side by side; rectal fistula commonest **within the vaginal septum**.
- **Early endoscopy**: distended vagina compresses bladder — **empty the vagina first**.

Common channel (Peña)	<3 cm	>3 cm
Anomalies; hydrocolpos	Fewer	More (urological defects nearly all)
Abdominal approach	Rarely	More likely
Bladder	Better spontaneous voiding	CIC far more often

5 · DSD reconstruction — timing and consent

- **For early surgery** (historically 2–6 months): pliable tissue, less trauma, parental relief. **Against**: advocacy/human-rights bodies seek a moratorium until consent; gender identity unknown.

- Most CAH patients/parents support parental decision; all families see a psychologist; observation always offered.
- **Clitoroplasty**: only for severe hypertrophy (**Prader III–IV**) — most controversial component.

High-confluence vaginoplasty	Arguments
Early single-stage	Redundant clitoral/sinus tissue usable; maternal oestrogen thickens tissue; planned one-stage does better
After puberty	Infant stenosis; avoids dilation; consent; delay the small vagina (**<3 cm**)
Hybrid	Clitorolabioplasty in infancy, vaginoplasty at puberty

KEY POINT

Both camps invoke oestrogen (maternal vs pubertal). Without comparative data the decision is ethical, not evidential; it rests with parents and team. Endocrine Society CAH guideline (2018): families with experienced surgeons decide; 2006 consensus limits surgery to Prader III–IV.

Body	Position
LWPES/ESPE CAH group (2002)	Surgery at 2–6 months; not between 12 months and adolescence
ISHID poll	Most prefer single-stage before 2 years
AMA ethics council	Rejected moratorium after 2 years' review
Advocacy groups; some US state bills	Moratorium until consent; bills not enacted

6 · Operative principles

Order: **clitoroplasty → vaginoplasty → labioplasty**.

Clitoral surgery	Technique	Verdict
Amputation; Lattimer recession	Excision; recession	Abandoned
Schmid (1961) MCQ	First corporeal excision preserving neurovascular bundle	Basis of later techniques
Kogan (1983)	Subtunical excision, lateral Buck fascia incisions	—
Baskin (1999)	Defined neuroanatomy	**Ventral** incision spares dorsal bundle
Pippi Salle (2007)	Corpora disassembled, buried in labia	Reversible if male identity emerges

- **Fortunoff (1964)** posterior perineal flap underlies almost every repair; now **omega-shaped**. **Hendren and Crawford (1969)**: pull-through; **confluence level determines vaginoplasty type**.

Vaginoplasty	Indication	Principle	Pitfall
Cut-back	Labial fusion only	Incise sinus	Misses normal-calibre vagina
Flap	Low confluence	Posterior sinus opened, flap inset; confluence unmoved	**Never for high confluence**: vaginal voiding, infection, incontinence

Vaginoplasty	Indication	Principle	Pitfall
Pull-through	Very high	Vagina separated; sinus becomes urethra	Vagina falls short → flaps
Replacement	Absent vagina	*→ Ch 58*	—

KEY POINT
Ask: **must the confluence be moved?** Flap brings skin to a low vagina; pull-through takes a high vagina off the urethra. Flap on a high confluence is the classic error.

7 · Low confluence repair

ALGORITHM
Metabolic stability → **stress-dose steroids in CAH** → antibiotics → endoscopy → **Fogarty in vagina** + Foley in bladder → total lower-body prep (supine–prone rotation; lithotomy blocks conversion)

ALGORITHM
Glans traction suture → incise around glans, keep inner prepuce for hood (**second only to glans in sensitivity**) → strips beside ventral mucosal plate → omega flap → Y-incisions on labia majora → deglove to bifurcation, ventral plate intact

- **Ventral subtunical excision**: incise Buck fascia on **ventral-most** corpora → tease out erectile tissue → tunics and dorsal bundle untouched → ligate stumps. Bundle mobilisation, if done, beneath second layer of Buck fascia.
- **Glans reduction**, if any, ventral midline; never excise glanular epithelium. Fix glans to corporal stumps, not pubis.

ALGORITHM
Flap vaginoplasty: omega flap with fat → posterior sinus off rectum (**hardest step**) → open posterior sinus in midline into posterior vaginal wall → **continue to normal-calibre vagina** → inset flap

Labioplasty: split clitoral skin as **labia minora** (Marberger); labia majora moved down by **YV-plasty**.

KEY POINT
Two fixed points: never touch the dorsal neurovascular bundle; always open back to normal-calibre vagina.

8 · High confluence: pull-through

- Only ≈ **5%** of CAH; commoner in pure sinus.
- **Critical step**: separating anterior vaginal wall from urethra/bladder neck — no plane, sphincter and nerves at risk → stricture, fistula, diverticulum.
- **Passerini-Glazel flap**: tubularised sinus + clitoral skin builds anterior vaginal wall. **Gonzalez–Fernandes**: preputial flap.
- **ASTRA** (anterior sagittal transanorectal): prone, anterior rectal wall divided — excellent exposure, no colostomy, continence unaffected.

ALGORITHM
Prone → midline between sinus and rectum → divide sinus to normal vagina → retractor inside vagina → anterior wall off urethra **under direct vision** → close sinus as urethra in 2–3 layers → mobilise vagina → supine: flaps + labioplasty

Err on the side of the vagina. Short vagina → skin flaps (sinus tissue more reliable). Very small, very high vagina: consider waiting until puberty.

9 · Total and partial urogenital mobilisation

- **TUM (Peña 1997)**: for cloaca; whole sinus mobilised circumferentially (dividing **avascular pubic ligaments**) to perineum without separation. Less fistula/stenosis, better cosmesis. **Enables** vaginoplasty rather than replacing it.
- **Never discard the sinus**: split **ventrally** = vestibule (flap repair); **dorsally** = anterior vaginal wall (pull-through).

	TUM	Partial (PUM, Rink)
Extent	Retropubic and suprapubic	Stops at **pubourethral ligament**
Use	Higher confluence; cloaca	Most sinus repairs
Risk	Sphincter/nerve injury, stress incontinence	Lower; convert to TUM if needed

10 · Outcomes of sinus/DSD surgery

- Infant vaginoplasty → **secondary introitoplasty** after puberty in **25–100%** (orifice does not grow); revision easy.
- **Azziz (1986)** MCQ: poor coitus rates, but almost all **cut-back** — not applicable now.
- Early single-stage → less stenosis than late; **confluence level, number of procedures, surgeon expertise** outweigh age. No data favour postpubertal surgery.
- Stenosis up to over a third (cut-backs); dyspareunia common. Orgasm same operated vs not. Sexual satisfaction as controls except **salt-wasting CAH**.
- **Elmasri meta-analysis (2018)**: mean age 2.7 years; female identity ≈ 89%; vaginal stenosis **27%**; only half comfortable intercourse; no early-vs-late comparison.

Two-procedure dilemma	Infancy	After puberty
Approach 1	Simple clitorolabioplasty	Extensive vaginoplasty
Approach 2	Extensive single-stage clitorolabiovaginoplasty	Simple introitoplasty

Both usually mean two operations. Key factor: **surgeon/team with special interest** → centres of excellence.

11 · Cloaca — initial management

- **Hendren's rule**: never a separate rectal pull-through; repair rectum, vagina and urethra in one stage. Some groups stage (anorectal first, urogenital peripubertally).

ALGORITHM
(1) Decompress bowel → (2) decompress genitourinary tract → (3) correct nephron-destructive anomalies → (4) definitive repair

Definitive repair is never the first operation.

Colostomy	Advantage	Drawback
Low loop	—	Bowel hard to obtain for vaginoplasty; interferes with pull-through
Right transverse, divided (Hendren)	Left colic supply intact; colon for vaginal replacement	Urine absorption → acidosis
Descending, divided (Levitt, Peña)	Less resorbing mucosa; enough distal colon; less prolapse	Leave a long sigmoid

- Colonic urine absorption → **hyperchloraemic acidosis**. **Preserve appendix** for later MACE.
- **Urinary decompression**: **CIC first line** (even into vagina) → cloacal cut-back → vaginostomy/vesicostomy only if CIC fails.
- **Kidneys**: structural anomalies **60–83%**; CKD **75%**, ESRD **17%**. Urodynamics abnormal in **90%**. Reflux managed medically if UTI prevented.

12 · Cloaca — definitive repair

At **6–12 months**, after urinary evaluation and exclusion of hydrocolpos.

Common channel	Urethra	Procedure (Wood algorithm)
<3 cm	**>1.5 cm**	**TUM**
>3 cm	or <1.5 cm	Urogenital separation
Any	Any	Transabdominal TUM never (ischaemia)

KEY POINT
Why 1.5 cm? Urethra >1.5 cm keeps the bladder neck above the urogenital diaphragm; mobilising a shorter one drags it below the continence mechanism → leakage.

ALGORITHM
PSARVUP (Peña): catheters in each channel → **prone jackknife** → muscle stimulator marks anal site → midline incision → open rectal fistula, mobilise rectum → TUM or vaginal separation → urethra in 2–3 layers + **fat flap** → perineal body → rectum centred in sphincter

- Rectum and vagina share a wall: **better to enter the vagina than rectum or urethra**. Rectum above pubococcygeal line → laparoscopy/laparotomy.
- Vaginal separation is tedious; vagina often **almost encircles the urethra**; **sharp dissection, not cautery**. Vaginal "switch" abandoned. Very high/absent vagina → bowel interposition *→ Ch 58*.
- Aftercare: Foley 2 weeks; rectal dilation from 2–3 weeks; colostomy closure ≈ 3 months. Poor emptying in almost one-third → CIC if any doubt.

13 · Cloaca — results and follow-up

- Outcome depends mostly on **spine** (short cord worst) and **common-channel length**.
- **Urinary**: about half void spontaneously; Peña: CIC **28%** (channel <3 cm) vs **78%** (>3 cm). Lifelong yearly review; LPP >40 cm H_2O common despite normal spinal MRI.
- **Renal**: CKD mostly stage I–II; preserved by aggressive bladder management.
- **Faecal**: tracks neurology; improves into adulthood. Bowel programme → **MACE**.
- **Gynaecological**: introital stenosis, retained septum, fistula, deficient perineal body; about half of outside repairs need redo; vaginal stenosis/obstructed menstruation **36–41%**;

pregnancy possible, **caesarean preferred**. Leave atretic genitalia.

ALGORITHM

US **6–9 months after thelarche** → repeat every 6–9 months until menses normal

65 Adolescent and Transitional Urology

1 · Definitions

- **Adolescent urology**: bridge for complex congenital conditions. Adult urologist has operative skill but lacks **whole-patient knowledge** (prior operations, altered anatomy, demanding families).
- **Transition** (Got Transition): planned move from child/family- to adult/patient-centred care — **planning → transfer → integration**. Transfer is one step.

> **KEY POINT**
> Evidence looks weak because the wrong endpoint is measured. Renal function and HbA1c are expected to worsen, so **stability and retention in follow-up** are success; fewer are lost when actively transitioned.

2 · Who needs transition

Category	Examples	Implication
1. Complex congenital	Exstrophy, PUV, complex hypospadias, spina bifida (SB)	Planned lifelong care
2. Treated, quiescent	VUR	Early-adult review; may discharge with open return
3. Cured, with sequelae	Pelvic rhabdomyosarcoma	Late-effects surveillance
4. New adult concern	Fertility, pregnancy	Adult expertise

Failed transition: poor understanding, overbearing family, loss to follow-up, ED reliance → higher morbidity and mortality via **lost preventive surveillance**.

3 · The process

"Three Ps"	Content	Success measure
Preparation	Independence from 12–14 years; TRAQ (readiness questionnaire)	All families involved
Plan	Clinic, imaging, bloods; self-care (catheterisation, irrigation, medications, whom to call)	Shared with patient and future team
Provider of the future	May or may not be a new team	Named provider

- **Paediatric provider must initiate**.
- Patient must be stable — no transfer during a crisis. Exception: **pregnancy** → expedited transfer with co-management.
- **Parents**: losing control is a major barrier. Give **private time** (sexual function), then ask what may be shared.
- Begin ≈ **12 years** (AAP); complete by late teens–early 20s; **26** = latest (US parental insurance). If independence unachievable, plan custodianship.
- Programmes: Good 2 Go (Toronto), Ready Steady Go (UK), Got Transition (USA).

Got Transition core element	Age (years)
Policy	12–14
Tracking, readiness, planning	14–18
Transfer of care	18–21
Transition complete (integration confirmed)	18–23

KEY POINT
Transfer is not the endpoint: transition ends only when integration into adult care is confirmed.

4 · Routes and models of care

- Routes: transitioned referral (ideal); referral without transition; **new adolescent problem (ideal)**; **emergency after discharge** (most challenging, often most prevalent).

Model	Strength	Drawback
1a Lifelong paediatric urologist	Continuity; suits limited life expectancy	Patient outlives provider
1b Paediatric urologist in adult department	Continuity + adult expertise	Cost, privileging, geography
2 Transfer to adolescent/adult specialist	Specialist coordinates MDT	Cross-institution communication
3 Direct transfer, no plan	Default	Invisible until emergency

Lifelong and multidisciplinary need are **independent**: exstrophy may need only a urologist; SB needs the most complex MDT. **Nursing** is the backbone.

5 · Development and consent

- Phallic–oedipal period (**3–6 years**) → avoid genital surgery; first year → avoid separation. Genital surgery: theory and 1975 AAP panel favour **<6 months**.
- Augmentation withheld until the child can catheterise and wants dryness.
- **Emerging adults (18–25)** consent for themselves unless lacking capacity; self-management matures into the third decade.
- **SB**: usually normal intelligence but deficits in **working memory and executive function**.
- **Medical PTSD**: distrust, avoidance, procedural panic; CBT may help.

6 · Setting up a service

- Core MDT: paediatric and adult urologists, specialist nurses, care coordinator, nephrologist, psychologist. Support via a **business model**.
- Consent and confidentiality reviewed **with family present, then alone**; standing orders (urine cultures) avoid ED visits.

Barrier	Solution
Patient/parent: attachment, fear	Address the child directly early; readiness questionnaires
Paediatric provider: family trust	Ask "Am I the best person?"
Adult provider: **unfamiliarity (biggest)**; finance	Partnerships, networks, training

120 Aging and Geriatric Urology

1 · Concepts and biology

- **Chronological age is a poor descriptor**: physiological and functional age (function, frailty, geriatric syndromes) matter.
- "Geriatrics" first used clinically by Nascher (1909) MCQ.
- **Cellular senescence** (irreversible replication arrest) drives ageing: replicative, via telomere shortening — **Hayflick limit** (1961) MCQ; or stress-induced (ROS, radiation, DNA damage).
- **SASP** (senescence-associated secretory phenotype): cytokines, growth factors, MMPs → chronic inflammation, tumorigenesis; ages neighbouring tissue.
- **Healthy ageing** (WHO): maintaining functional ability — target function, not absence of disease.

2 · Clinical evaluation

- Geriatric syndromes act **synergistically** (many factors → one phenotype). **Homeostenosis**: narrowed physiological reserve, poorer recovery after a stressor.

Tier	Content	Significance
Basic ADLs	Bathing, dressing, grooming, toileting, feeding, ambulation	**Bathing lost first**, feeding last; loss predicts dependency and postoperative outcome
Instrumental (IADLs)	Cooking, cleaning, transport, finances, telephone, medicines	Independent living

- Function lost top down; each step shortens survival. Screen: **can the patient bathe independently?**
- **Surgical risk**: age seldom independent; predictors are **longer operating time, poor ASA score, urgent surgery**. Optimise, do not "clear" — benefits even >80.

> **ALGORITHM**
> Frailty screen before elective major surgery (SPAQI) → positive → diagnostic frailty assessment → comprehensive geriatric assessment (CGA)

- **ACS NSQIP/AGS checklist**: 13 items (cognition/capacity, depression, delirium risk, substance use, cardiac and pulmonary risk, function/falls, frailty, nutrition, medications, goals, social support, tests); only 2 organ systems.
- **CGA**: geriatrician-led — frailty, physical performance, albumin, creatinine, BNP, FEV1.

3 · Geriatric syndromes

- Frailty, depression, delirium, falls, incontinence, polypharmacy, sensory loss, pressure ulcers — feed one another (frailty ↔ UI ↔ falls), so treating one helps others.

Frailty = decreased reserve → vulnerability; distinct from comorbidity and disability.

	Accumulated deficits index	Fried phenotype
Method	Count failing systems	(1) weight loss >4.5 kg or 5%/year; (2) weak grip; (3) slow gait; (4) low activity; (5) exhaustion
Cut-off	Continuous	**≥3 frail; 1–2 pre-frail**

	Accumulated deficits index	Fried phenotype
Validity	Does not predict postoperative outcome	Predicts death, dependence, falls, postoperative complications

Polypharmacy — Beers criteria

Harm often via the **prescribing cascade** (side effect treated with another drug).

Drug	Concern → action
Antimuscarinics (dementia/delirium), tricyclics, first-generation antihistamines	Anticholinergic burden: delirium, falls, retention → avoid combinations; no tricyclics in BPH
NSAIDs	GI bleeding; renal and heart failure at CrCl <30 mL/min
Systemic oestrogen; testosterone	Cancer, CV harm → **vaginal oestrogen safe**; testosterone only for confirmed symptomatic hypogonadism
Ciprofloxacin, macrolides (**not azithromycin**), co-trimoxazole	Warfarin potentiation → monitor INR; dose-reduce at CrCl <30
Benzodiazepines	Cognition, gait, delirium → avoid with opioids or ≥2 other CNS drugs
Nitrofurantoin	Listed at CrCl <30/chronic use; usable in moderate impairment
Desmopressin; tramadol	Hyponatraemia → treat nocturia's cause; check sodium

> **KEY POINT**
> Four recurring mechanisms: anticholinergic burden, renal threshold (CrCl <30), warfarin potentiation, sodium disturbance.

Genitourinary syndrome of menopause (GSM)

- Coined 2014; transient-incontinence cause. Dryness, dyspareunia, UTI **plus urgency and UUI**; sex-steroid loss drives OAB in both sexes (incl. ADT, 5-ARIs).
- **Vaginal oestrogen** (or moisturisers, DHEA, ospemifene); prior breast/endometrial cancer not an absolute bar — consult oncologist.
- **Urethral caruncle**: painless spotting — exclude endometrial/cervical cancer; topical oestrogen, excision rarely.
- Faecal incontinence under-reported — **ask specifically**.

4 · Incontinence management

- Common but not normal; behavioural therapy first → *Voiding Dysfunction book, Ch 110*. †Transient causes (DIAPPERS): delirium, infection, atrophic vaginitis, pharmaceuticals, psychological, excess output, restricted mobility, stool impaction.
- **Antimuscarinics**: < half persist at 6 months; leakier blood–brain barrier → cognitive harm. Check PVR after starting; never exceed maximum dose.
- P2 P3 Overactive bladder and its drug therapy; OAB Pharmacotherapy — *trospium quaternary, M3 selectivity, tolterodine QT, mirabegron BP/metoprolol, Beers avoid in dementia*
- Beers "avoid" in dementia: **oxybutynin, fesoterodine**; trospium least central penetration.

Nursing homes

- UI in three-quarters of long-stay residents; drives admission. Catheter colonisation **98%** → treating ASB harms.

Loeb — catheter	Loeb — no catheter	NHSN
≥1: fever, new costovertebral tenderness, rigors, **acute mental status change**	Acute dysuria; or fever + new urgency, frequency, suprapubic pain, haematuria or UI	Culture ≥10^5 CFU + no other source + clinical signs

KEY POINT
A positive culture alone never qualifies; catheterised residents cannot report local symptoms, so systemic signs are used.

5 · UTI and asymptomatic bacteriuria

- Separate UTI from OAB and ASB. **Confusion–UTI causal link not supported** *→ Infections book, Ch 26*.
- ASB: up to 22% older women, ~10% older men; no antibiotics. USPSTF grade D against screening; IDSA: treat **only in pregnancy and before urological instrumentation**.

6 · Cancer, end of life

- **Prostate**: USPSTF grade D — no PSA screening >70; life expectancy **<10 years** → non-curative plan.
- **Bladder**: restaging TURBT for T1 often omitted in elderly yet improves survival; BCG less effective (immunosenescence); cystectomy at ≥75 comparable with careful care.
- Urologists are mandatory reporters of **elder mistreatment** in most places.
- **Palliative care ≠ withdrawal**: aggressive symptom control; goals-of-care first.
- Age-Friendly "4 Ms": Mobility, Medication, Mentation, what Matters.

KEY POINT
Undertreatment by age (skipped restaging, withheld cystectomy) is as real a risk as overtreatment. Fitness and life expectancy decide.

7 · Landmark trials

Trial	Design	Result	Take-home
ASPREE	RCT, aspirin vs placebo, healthy elderly	No disability-free survival gain; higher mortality	No primary-prevention aspirin
Testosterone Trials	RCTs, hypogonadal older men	No cognitive benefit; modest sexual-function gain	Not for cognition

Effective prevention: calcium + vitamin D (fractures), antihypertensives (mortality even >80), statins, exercise.

126 Management and Considerations in Transgender and Nonbinary Individuals

1 · Definitions and terminology

- **Transgender and gender-diverse (TGD)**: gender identity differs from sex assigned at birth. Four **independent axes** — assigned sex, gender identity, gender expression, sexual orientation; most errors collapse one into another.

Term	Meaning
Gender identity / expression	Internal sense of gender / its outward display
AFAB / AMAB	Assigned female/male at birth (genitalia and/or genetics)
Cisgender	Identity congruent with assigned sex
Transgender man / woman	Male identity, AFAB / female identity, AMAB
Transmasculine / transfeminine	Direction of gender, not a binary endpoint
Nonbinary; agender; gender fluid	Neither male nor female; no gender; changes over time
Gender dysphoria	Distress from identity–assigned sex discrepancy
Gender incongruence	Diagnostic term: marked, persistent incompatibility
Sexual orientation	Attraction — not inferable from identity
"Transsexual"	Historical; disfavoured

- Not all TGD people have dysphoria; some need only social transition, others GAHT and/or gender-affirming surgery.

2 · Diagnosis and assessment

	DSM-5 (2013)	ICD-11 (2019)
Term	Gender dysphoria	**Gender incongruence** (replaced "gender identity disorder")
Criteria	Marked incongruence ≥ **6 months**, significant distress or impairment, not solely due to prejudice	Marked, persistent incongruence; desire for characteristics/treatment of experienced gender
Chapter	Mental disorders	**Moved to Conditions Related to Sexual Health** MCQ

- ICD-11 keeps a code for access, drops mental-disorder label. **Severe distress not a prerequisite** — marked, sustained incongruence is.

WPATH Standards of Care 8 (SOC-8, 2022)

- Treat only marked, sustained incongruence; exclude other causes; optimise coexisting conditions.
- Assess **capacity to consent**; **fertility counselling before any treatment**, not just surgery.
- **A single opinion** from a competent professional suffices for medical and surgical treatment.
- Before gonadectomy **consider ≥6 months of hormones** unless not indicated (a consideration, not a rule).

KEY POINT

SOC-8 replaces gatekeeping with informed, capacity-based consent; capacity and fertility counselling stay firm because consequences are irreversible.

3 · Gender-affirming hormone therapy (GAHT)

- Safe under supervision; aim: **physiological** levels of the affirmed gender. Eligibility: persistent documented dysphoria, capacity, age of majority, controlled comorbidities.

Direction	Agent	Dose
Transfeminine — oestrogen	Oral oestradiol	2–6 mg/day
	Transdermal oestradiol	0.025–0.2 mg/day
	Oestradiol valerate/cypionate IM	5–30 mg 2-weekly or 2–10 mg weekly
Transfeminine — antiandrogen	Spironolactone	100–300 mg/day
	Cyproterone acetate	25–50 mg/day
	GnRH agonist	3.75 mg monthly / 11.25 mg 3-monthly
Transmasculine — testosterone	Enanthate/cypionate	100–200 mg 2-weekly
	Undecanoate	1000 mg 12-weekly (rare in US: oil pulmonary embolism)
	Gel 1.6% / patch	50–100 mg/day / 2.5–7.5 mg/day

KEY POINT

Why the asymmetry? Physiological oestrogen alone does not suppress testosterone to female range → transfeminine therapy needs two components; testosterone suppresses the ovarian axis unaided → one.

Transfeminine

- **Ethinyl oestradiol not recommended** (~20-fold thrombosis; unmeasurable). Oral 17β-oestradiol, target **100–200 pg/mL**. **Transdermal if VTE risk**: age >45 or previous VTE.
- **Spironolactone** (US commonest): AR antagonist + partial 17α-hydroxylase/17,20-lyase inhibition. **Cyproterone**: European standard. Alternatives: GnRH agonists, bicalutamide. 5-ARIs weaker; progesterone unproven.
- Effects 3–12 months. **Irreversible: breast development, reduced/absent fertility.** Risks: VTE, CVD, hypertriglyceridaemia, liver dysfunction, osteoporosis, hormone-related cancer.
- Monitor levels 3-monthly in year 1, then 1–2×/year; target testosterone **<50 ng/dL**; electrolytes/creatinine on spironolactone; prolactin; stop smoking.
- **After gonadectomy: reduce oestrogen by 50–75%** (titrate to LH/FSH, bone density) and **stop the antiandrogen**.

Transmasculine

- Contraindications: pregnancy, unstable coronary disease, untreated polycythaemia. Oestrogen-sensitive cancer → oncology (aromatisation). **Contraception essential** — testosterone teratogenic.
- Target testosterone **300–1000 ng/dL**. Persistent menses → progestin or endometrial ablation.

- 1–2 months: menses stop, acne, clitoromegaly, vaginal atrophy, libido up; later hair, muscle, voice. **Irreversible: clitoral growth, facial hair, voice change, male-pattern baldness.** Menses return 2–6 months after stopping.
- Risks: **erythrocytosis**, liver dysfunction, sleep apnoea, CVD. Monitor testosterone and **haematocrit** 3-monthly in year 1, then 1–2×/year.
- **After gonadectomy: no testosterone dose reduction**.

Nonbinary: individualised — AFAB: progestin/IUD/hysterectomy + testosterone and/or **nandrolone** (muscle without much hair); AMAB wanting preserved erections: oestrogen ± low-dose antiandrogen.

4 · Urological considerations

LUTS contributors: **natal anatomy, hormones, surgery, pelvic floor dysfunction, psychosocial/behavioural**.

- **Testosterone-induced vaginal atrophy**: dryness, bleeding, dyspareunia, urgency, frequency → topical vaginal oestrogen.
- **After vaginoplasty**: ~a third altered voiding (**nocturia commonest**); spraying; transient incontinence; meatal stenosis, stricture; urethroneovaginal fistula → *Ch 127*.
- **After phalloplasty/metoidioplasty**: post-void dribble from pooling at native urethra–pars fixa junction (compress scrotal base); stricture and fistula far commoner → *Ch 128*.
- **Pelvic floor**: treat before surgery; eases neovaginal dilation.
- **Tucking**: stasis, prostatitis-like symptoms, epididymo-orchitis, UTI.
- Exclude atrophy, tucking, pelvic floor causes before blaming the operation.

5 · Fertility preservation

- Counsel **before hormones and/or surgery** (WPATH, ACOG, ASRM, Endocrine Society). Barriers: cost, urgency, bias, process-provoked dysphoria.

	Transfeminine	Transmasculine
Standard	**Sperm cryopreservation**; hold GAHT ≥ **3 months** before collection	**Oocyte or embryo cryopreservation** after ovarian stimulation
Prepubertal / no pause	Testicular tissue cryopreservation (experimental)	Ovarian tissue — no hormone pause or stimulation
Obstacles	Ejaculation distress → vibratory/electroejaculation or surgical retrieval	Bleeding returns; pelvic exams/TVUS distress

Off testosterone, menses return in 3–6 months; most conceive within 6 months.

6 · Cancer screening — screen the organ, not the gender

- **Prostate (transfeminine)**: rare, may be aggressive (GAHT after 50). Shared decision on PSA at 55–69. **On GAHT, upper limit 1 ng/mL or rise ≥0.3 ng/mL from nadir** †(androgen suppression lowers PSA). After full-depth vaginoplasty, palpate via **anterior neovaginal wall**; biopsy transrectal, perineal or transneovaginal.
- **Testis**: only case reports — self-examination only.
- **Cervix (transmasculine)**: cytology 3-yearly at 21–29; 30–65 cytology 3-yearly or hrHPV/co-testing 5-yearly; stop if no cervix or >65 with two negatives. Small speculum, **vaginal**

oestradiol 1–2 weeks before.

7 · Affirming practice

- Lifelong organ-appropriate screening; **detailed organ inventory**; inclusive intake (name, pronouns).
- **Trauma-informed care** (SAMHSA) "Four Rs": **Realise, Recognise, Respond, Resist retraumatisation**.

127 Male-to-Female Gender-Affirming Surgery: Vaginoplasty

1 · Aim and preparation

- Align anatomy with identity **and** preserve voiding, sensation, sexual function. Terminology, hormones → *Ch 126*.
- **Mental health**: treat dysphoria and judge **surgical readiness**; involve early (housing, support, smoking, weight). Key risks: **failure to dilate**, worsening psychiatric prognosis. Dysphoria-driven depression/suicidality is expected to **improve** — not a reason to defer.

Older criteria	WPATH SOC8 (informed-consent approach)
Two assessments	**One** (high inter-assessor agreement)
Mental health professional only	Any licensed clinician ≥ master's degree, competent in transgender health (incl. primary care or surgeon)
Hormones before gonadectomy	**6 months** suffices (experience testosterone blockade before irreversible change)
Comorbidity "well managed"	Risks and benefits discussed
12 months' "real-life experience"	**Removed** — dangerous for those facing violence

- **Capacity is procedure-specific**; if doubtful, psychologist plans support ± proxy decision-makers.

> **KEY POINT**
> SOC8 replaces thresholds to clear with information to receive; capacity and fertility counselling stay mandatory.

- Stay within **1 hour's drive** of hospital for the first **3 weeks**; record affirmed name/gender; confirm support for dilation.
- **BMI**: main factor in wound healing; no proven cut-off, many require **≤35 kg/m²**; higher may suit vulvoplasty.
- **Permanent preoperative hair removal** of scrotal–perineal graft template (intraoperative follicle removal fails — phased growth). Canal hair → infection, hairballs.
- **Continue oestrogen perioperatively**: 2022 meta-analysis — no added VTE risk.

Orchiectomy

- Sole or first stage; afterwards essentially all stop spironolactone. **Fertility counselling and sperm banking for all**.
- **Vertical midline scrotal incision**: inguinal may compromise penile flap supply; scrotal incision is incorporated into the later graft.

> **KEY POINT**
> Plan every orchiectomy as though vaginoplasty will follow — midline scrotal incision, peri-cord fat left for labial bulk.

2 · Penile inversion vaginoplasty (PIV)

- P2 P4 Penile inversion vaginoplasty technique — *homologous tissue table, operative steps, dilation, complication rates*
- Set-up: lithotomy, pad calf/heel (**common peroneal nerve**); heparin + compression; **tranexamic acid 1 g IV at start** (front-loaded bleeding).

ALGORITHM

Foley, drain → mark adductor longus tendons (clitoral level) and dorsal midline (prevents torsion) → scrotal graft + small posterior rhomboid flap (relieves introital tension) → defat, tubularise → orchiectomy, leave cord fat → disassembly → corporal excision + clitoroplasty → urethroplasty → canal dissection → line canal → labiaplasty → pack

- **Disassembly**: keep dartos with skin; neurovascular bundle best **left on the dorsal tunica** (less nerve damage); open corpora ventrally, coagulate cavernosal arteries. **Excise all erectile tissue**.
- **Clitoroplasty**: "W" on proximal glans, keep lateral wings; too much glans → penis-like clitoris. Short penile shaft shortens labia minora, not canal.
- **Urethroplasty**: excise bulbospongiosus **only after canal dissection** — it covers a rectal injury. Spatulate downward for a seated stream.
- **Canal**: divide central tendon; Lowsley retractor; repeated DRE. **Preserve Denonvilliers fascia** (close if breached). Prostatic capsule breach → check urethra. **Packing** ensures graft take.
- Labia majora incision **≥1 cm** from groin crease (dehiscence). Antibiotic pack + negative-pressure dressing.

Vulvoplasty (zero-depth)

- For those not wanting receptive intercourse: avoids **lifelong dilation**, rectal/bladder injury. Not freely reversible (graft discarded, scarring).
- **Prior radical prostatectomy → vulvoplasty** (sphincter injury risk).

3 · Abdominoperineal and intestinal options

- **Peritoneal (Davydov-type)**: proximal canal made robotically; rectal injury **~7%** historically → ~2.3% (urethral 1.1%). **Rectum may mimic seminal vesicles**; enter between Denonvilliers leaflets.
- Flaps: **two-flap** (bladder + sigmoid peritoneum, 10 × 10 cm each) or **pull-through** (abdominal wall 11 × 20 cm on deep inferior epigastrics). **Close peritoneal defects** (hernia).
- **Intestinal**: severe penoscrotal hypoplasia or revision stenosis. CI: IBD, adhesions, irradiation, short gut. Mucus, diversion colitis, prolapse; ~1 in 8 removed. **Dilation still required.**

KEY POINT

Lining ladder: penile/scrotal skin → peritoneum → bowel. Each adds depth and lubrication but imports donor pathology; none removes the duty to dilate.

4 · Complications

Complication	Cause / key fact	Management
Wound dehiscence (commonest minor)	Introitus, dilator tension	Local care; granulation → curette/cautery + steroid
Rectovaginal fistula (most devastating)	**0.8%** primary, 6.3% revision	Intra-op: watertight closure + bulbospongiosus flap ± diversion. Fistula: stop dilating, let canal stenose, revise later **or** delay closure
Urethrovaginal fistula (0.8–3.9%)	Missed injury, distal stenosis	Distal (vaginal voiding) often left; bladder-neck (incontinence) → transvaginal repair; Martius unusable
Urethral stricture (1–6%)	Split stream, spraying, UTI, retention	Exclude pelvic floor; dilation if short, else meatal spatulation
Urethral bulge / upward meatus	Under-debulked spongiosum	Debulk, fix meatus down
Vaginal stenosis	Poor dilation, scar, graft loss, pelvic floor	Pelvic floor physiotherapy **first** → scar excision + graft, or peritoneal/bowel augmentation
Prolapse (4–10%)	Poor graft take/fixation	Early repack; peritoneal fixation acts as colpopexy

KEY POINT

Three failure modes: violated plane (fistulas), tissue not excised (bulge, engorgement), graft not taken/maintained (dehiscence, stenosis, prolapse). Only the last depends on the patient — dilation is part of the operation.

5 · Outcomes and follow-up

- **Median time to orgasm ≈ 180 days**. Depth 12–16 cm (cisgender 9–12 cm); self-lubrication not the norm.
- Stay ~**5 days**; **day 5: pack and catheter out**, teach dilation. Retention → recatheterise 1 week.
- Follow-up: weekly × 3, then 6 weeks, 3, 6, 12 months, annually. †Prostate retained — surveillance via anterior neovaginal wall.

Period	Dilation (15–20 min each)
0–6 weeks	4 × daily
6 weeks–3 months	3 × daily
3–12 months	2 × daily
>1 year	Daily; less if stable

KEY POINT

The body treats the canal as a defect to close — lifelong stenosis risk. Any narrowing → more dilation, contact surgeon, physiotherapy.

128 Special Urologic Considerations in Transmasculine Individuals

- P4 Enlist the steps in female to male gender reassignment surgery — *staged sequence, vaginectomy, neourethra parts, metoidioplasty vs phalloplasty, scrotoplasty, prostheses, complication sites*

1 · Scope

- Diagnosis, WPATH, hormones *→ Ch 126*. Genital surgery inevitably affects the urinary tract; goals: male appearance, **voiding standing**, sensation, penetration.
- **Early puberty blockade has a surgical price**: oestrogen-naive vagina → harder vaginectomy; small labia majora → less scrotal tissue. Avoiding public toilets may cause underactive detrusor.

2 · Preoperative care and consent

- **Mandatory urological screening** (uroflow + PVR, renal US); treat LUTD, stones, hydronephrosis first — far harder after lengthening.
- WPATH SOC7: ≥**12 months** testosterone before genital surgery (†SOC8: 6 months) — for **clitoral hypertrophy** (plateaus ≈2 years).

Agent	Helps	Hinders
Testosterone	Clitoral hypertrophy → metoidioplasty, lengthening	Vaginal atrophy → harder vaginectomy
GnRH agonist	Prevents female characteristics	Small labia majora → less scrotoplasty tissue

- Consent: fistula/stricture; vaginectomy bleeding, mucocele, rectal/bladder/sphincter injury; LUTD, UTI; implant problems.

3 · Removal of female structures

- Ureter beneath uterine artery → injury needs immediate repair. †Retained cervix → screening continues.
- **Vaginectomy**: haemorrhage > bladder > rectal injury. **No vaginectomy → no urethral lengthening** (fistula). Alternative: fulgurate upper vaginal mucosa (prevents mucocele).

ALGORITHM

Hydrodissect with dilute lidocaine → circumferential introital incision → blunt rectovaginal plane → **sharp urethrovaginal dissection, no diathermy** (sphincter) → bladder plane; injury → close ± cystoscopy → free levators, control paravaginal plexus → approximate levators (prolapse) → bulbospongiosus over urethra → layered closure

Timing	Verdict
All-in-one	More major complications/transfusion with phalloplasty; none extra with metoidioplasty
Early vaginectomy, phalloplasty later	Keeps blood loss away from free-flap perfusion (theoretical)
Two-stage (preferred)	Mastectomy + hysterectomy–oophorectomy; then vaginectomy + genitoperineal reconstruction

4 · Genital reconstruction

- Fixed (perineal) urethra = male bulbar without spongiosum; phallic = penile without spongiosum. One **dorsal clitoral nerve** (1 or 11 o'clock) coapted for sensation.
- **Bulbospongiosus over the fixed part**: fewer fistulas, expels last drops. Anterior vaginal wall flap avoided (necrosis, sacculation).
- Scrotoplasty flaps on **external pudendal artery**. Phalloplasty: many flaps, none ideal.

	Metoidioplasty	Phalloplasty
Phallus	Own clitoris, **3–8 cm**	Flap
Erection	Spontaneous	Needs prosthesis
Sensation	Largely kept	Depends on nerve coaptation
Penetration	Usually not	Principal goal
Donor scar	None	Visible

5 · Postoperative care and catheterisation

- SPC + neourethral catheter; neourethral out at **14–21 days**; clamp SPC, record PVR until voiding normal.
- Catheterisation hard: abrupt curve into native urethra; Bartholin-duct sinuses catch the tip.

> **ALGORITHM**
> One gentle Coudé attempt → stylet-guided attempt → any difficulty: stop → flexible cystoscopy, guidewire, open-tip catheter. Emergency: **suprapubic catheter**

6 · Urethral complications

- **Fistula** (commonest at phallic base): keep diversion; small ones close; persisting >**3 months** → excise, layered closure with fat interposition, catheter 10–12 days.
- **Stricture** (commonest at **fixed–phallic anastomosis**; also meatus): plateau uroflow → RUG. Treat only after **≥6 months**; retention → SPC or perineal urethrostomy.

Option	Use
Dilation	Low-grade meatal stenosis; not curative for anastomotic
DVIU	Cold knife over guidewire, catheter 2 weeks; low success, never repeat
Meatotomy / Heineke–Mikulicz	Short meatal / short non-obliterative
Excision and primary anastomosis	Short, near-obliterative anastomotic
Free graft	Needs flap support (fat bed) or dorsal inlay on clitoral corpus
Two-stage (standard)	Long/complex; retubularise ≥6 months later
Flap / perineal urethrostomy	Recurrent; or no further reconstruction wanted

- **Hair**: hairless flap or preoperative laser; bezoar/stones → open, remove, epilate, retubularise later.
- **Post-void dribbling**: near-universal (no spongiosum) → perineal pressure, milking; severe → exclude sacculation or vaginal-remnant diverticulum.

KEY POINT

The neourethra is skin in fat without spongiosum: fistulas and strictures cluster at poorly perfused flap ends, heal slowly and resist endoscopic cure.

7 · Prostheses

- Erectile: ≥ **1 year** after phalloplasty, with **protective sensation** and stable urethra; one cylinder anchored to pubic periosteum (no tunica). Malleable simpler.
- Infection/erosion frequent → urgent explantation, reimplant ≥6 months.

★ Putting it together

A · Newborn with atypical genitalia

ALGORITHM

1. Emergency is adrenal, not genital: CAH commonest (half of all, 90% of 46,XX). Salt wasting (low Na, high K, shock) appears day 5–7 (→ *Ch 59*).

2. Tests: karyotype (SRY), 17-OHP after 36 h, electrolytes, pelvic US for uterus.

3. Team: †defer sex assignment until MDT evaluation.

4. Gonads: dysgenetic with Y (e.g. complete gonadal dysgenesis, 30–45% tumour risk) → prophylactic gonadectomy.

B · Urogenital sinus and cloaca

KEY POINT

True urethral length (bladder neck → confluence) outweighs common-channel length. Low confluence → flap; high → pull-through or TUM/PUM. Clitoroplasty only Prader III–IV, sparing dorsal bundle. Cloaca: colostomy, decompress hydrocolpos, cloacogram/MRI + endoscopy, then posterior sagittal repair; seek VACTERL, tethered cord (→ *Ch 60*).

C · Absent vagina

Cause	Key feature	Management
MRKH	46,XX, normal ovaries/breasts, no uterus; †image kidneys	Dilation first; else McIndoe, bowel, buccal, peritoneal
CAIS	46,XY, testes, no uterus	Dilation usually suffices (→ *Ch 59*)
Transverse septum / imperforate hymen	Uterus present, pubertal haematocolpos	Hymen: transverse incision, not aspiration; septum: excision (→ *Ch 58*)

D · Immunotherapy and hereditary cancer

ALGORITHM

Checkpoints: tumour IFN-γ induces PD-L1, silencing PD-1+ T cells; CTLA-4 inhibitors act earlier (priming). †irAEs (colitis, hepatitis, pneumonitis, endocrinopathy): hold drug + steroids for grade ≥2; endocrinopathies need lifelong replacement (→ *Ch 24*).

Hereditary: VHL (clear cell; belzutifan), HLRCC (FH; aggressive papillary), Birt–Hogg–Dubé (FLCN; hybrid oncocytic), Lynch (MMR; UTUC). BRCA/HRR prostate → PARP inhibitor; FGFR urothelial → erdafitinib (→ *Ch 25*).

E · Older patient before surgery

KEY POINT

Function, not age, predicts outcome. Frailty screen → CGA: cognition, delirium, nutrition, falls, polypharmacy. Avoid anticholinergic burden (β3-agonist for OAB). New incontinence: transient causes first (†DIAPPERS) (→ *Ch 120*).

F · Transgender and nonbinary patients

ALGORITHM

Organ inventory guides screening — retained prostate after vaginoplasty (anterior neovaginal wall; †PSA lower on oestrogen), retained cervix on testosterone (→ *Ch 126*).

Vaginoplasty: stenosis, rectovaginal fistula (most devastating), urethrovaginal fistula (→ *Ch 127*). Phalloplasty: fistula, stricture; no vaginectomy → more fistula (→ *Ch 128*).